

Spiral Fibonacci sequences (OEIS A094926, A094925 and A078510)

Part I: hexagonal spiral Fibonacci asymptotics, certified amplitudes
and all fixed order corrections

Part II: square Spiro-Fibonacci growth and inversion
Two independent studies of one construction family

Two research reports of one external research session
(bundle Reports 190 and 191), bound into one ProveIt report

Both Parts: 4 October 2026 merged: 7 October 2026

Abstract

This report binds two independent manuscripts on two sequences of the spiral-Fibonacci family, in which numbers are written along a planar spiral and each new cell adds values at earlier neighbouring cells. The Parts share this construction family and a method, but no theorem: neither uses, implies or re-proves the other. Part I treats the hexagonal spiral sequences A094926 and A094925. It reduces the geometry to an exact recurrence with six-side delays of order $\sqrt{n}$, proves $a_n \sim C\phi^n$ for both seeds, which proves the equivalents conjectured by Manfred Scheucher in 2015 in both OEIS entries, certifies at least 120 digits of the amplitudes by exact algebraic bounds, expands $a_n/(C\phi^n)$ to every fixed stretched-exponential order with nonzero sector profiles, proves the sharp deficit $\sqrt{n} e^{-2\sqrt{3}\log(\phi)\sqrt{n}} G(\{\sqrt{12n}\})$ with exact envelopes, and brackets the integer inverse. Part II treats the square Spiro-Fibonacci sequence A078510, whose recurrence $a_n = a_{n-1} + a_{t(n)}$ has the delay $n - t(n) = 4\sqrt{n} + O(1)$. It proves $a_n = Ce^{H(n)}(1 + O(w^3/\sqrt{n}))$ with $w = W_0(4\sqrt{n})$ and $C > 0$, by a periodic Bernoulli normalizer and a common-anchor contraction, inverts the threshold with absolute error $O(w^2)$, and gives every fixed inverse-logarithmic order. The write adds a proof that Scheucher's accompanying heuristic form $a(n) = (a(n-1) + a(n-2))/(1 - cd^{-\sqrt{n}})$ cannot hold with constants, and places the inverses relative to the repository's transseries volume. The report is unrefereed, and nothing in it is formalized.

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Guide to this report

The two Parts are two manuscripts of the session bundle of Reports 1–243 (arrival commit 60f54ea06), placed in this directory by commit 8622ca7e5 (batch 110, cluster 110-SPIRAL).

Part	Bundle report	Delivered title	Labels
I	190 (base)	Hexagonal Spiral Fibonacci Asymptotics: Certified amplitudes and all fixed order feedback corrections (4 October 2026)	sfb:hex:
II	191	Square Spiro Fibonacci Growth and Inversion: Positive amplitude and quantitative asymptotics for A078510 (4 October 2026)	sfb:sq:

Two independent Parts. This report is two self-contained studies of two different sequences that belong to one construction family, the spiral-Fibonacci (“Spiro-Fibonacci”) sequences in the sense of Fernandez [5]: numbers are written along a planar spiral and each new cell adds values at earlier neighbouring cells. The Parts share that family and a method, and nothing else. **No theorem, lemma or constant of one Part is used in, implied by, or a special case of the other**, and neither Part re-proves anything of the other; each is complete with its own definitions, hypotheses and proofs, and is printed in full. They are bound in one report because they are the repository’s only treatment of spiral-Fibonacci recurrences, because Report 191 calls Report 190 its “prior companion report”, and because their methods run in parallel; the merge claims no more than that. The table shows how far the parallel goes.

	Part I (hexagonal, Report 190)	Part II (square, Report 191)
Lattice and rule	triangular lattice; add all earlier neighbours of the preceding vertex (Kohmoto)	square cells; add the two closest earlier centres (Fernandez)
Sequences	A094926, A094925 (two seeds)	A078510
Exact recurrence	$a_n = a_{n-1} + a_{n-2} + \sum_{j \in E_n} a_j$, extra delays $6r - 4 + t$, $r \approx \sqrt{n/3}$	$a_n = a_{n-1} + a_{t(n)}$, $n - t(n) = 4\sqrt{n} + O(1)$
Growth	$a_n \sim C\phi^n$, deficit $\sqrt{n}e^{-2\sqrt{3}\log(\phi)\sqrt{n}}$	$a_n = Ce^{H(n)}(1 + O(w^3/\sqrt{n}))$, $\log a_n \sim \frac{1}{4}\sqrt{n} \log n$
Amplitude	exists; 120 certified digits	exists, $C > 0$; no digits
Expansion	every fixed stretched-exponential order	one relative error term only
Mechanism	positive Perron projection; variation of constants	periodic Bernoulli normalizer; common-anchor contraction
Inverse	two-ceiling envelopes at every fixed order	absolute error $O(w_0^2)$; inverse-logarithmic hierarchy

Report 191 says so itself: Part I’s theorem “cannot be obtained by changing a geometric constant” into its own (Section 21). The square rule has no term a_{n-2} , so its growth is subexponential, not $C\phi^n$.

Arrangement. The base of the merge is Report 190; its `Report190.tex` was staged as this `article.tex`, and it is printed first, as Part I, because Report 191 cites it as prior and because it settles two recorded OEIS conjectures. Report 191 is printed second, as Part II. Nothing is printed twice.

Numbering. Sections are numbered continuously; statements are numbered within sections and equations continuously through the report, as in both manuscripts. Part I keeps Report 190’s Sections 1–11, its equations (1)–(66) and all its other numbers. Report 191’s Section k is Section $k+11$

here (Sections 12–22), its statements move with their sections (Report 191’s Theorem 1.1 is Theorem 12.1), and its equation (k) is equation $(k + 66)$. Section 23, which gathers the open questions of both Parts under Vladimir’s standing rule of 4 October 2026, was added in the write.

Notes added in the write are set in small type and tagged **[write]**, with their date (7 October 2026). No delivered statement was changed and no delivered symbol was renamed.

What is proved, and where

Statement	Where
<i>Hexagonal spiral, A094926 and A094925 (Part I)</i>	
Six-side delay formula for the exact recurrence	Proposition 2.1
$a_n \sim C\phi^n$ for both seeds: Scheucher’s conjectured equivalents in A094926 and A094925 are proved ; every fixed stretched-exponential order	Theorem 1.1, Sections 4–6
Algebraic certificates for $C_0, C_1, C_1/\phi$ to at least 120 truncated digits	Theorem 4.1
Seed independence beyond every fixed order; nonzero leading profile of every sector	Corollary 6.2, Theorem 7.1
Sharp square-root phase $G(\{\sqrt{12n}\})$ with exact envelopes	Theorem 1.2, Section 8
The uncorrected logarithmic ceiling misses by one infinitely often; two-ceiling inverse envelopes at every fixed order	Theorems 9.1 and 9.2
<i>Square spiral, A078510 (Part II)</i>	
Occupied rectangles and the exact predecessor $t(n)$	Lemma 13.1, Proposition 13.2
$a_n = Ce^{H(n)}(1 + O(w_n^3/r_n))$ with $C > 0$, uniformly including corners	Theorem 12.1
The smooth normalizer has non-summable defects; the contraction product	Propositions 17.1 and 17.2
Threshold inverse with absolute error $O(w_0^2)$; every fixed inverse-logarithmic order	Theorems 12.2 and 19.2
<i>Open</i>	
Part I: behaviour as the order grows, closed forms of the amplitudes, effective onsets, a uniform seed theorem. Part II: certified digits of C , a sharper relative error, further sequence corrections, effective threshold recovery, general delayed models. See Section 23.	

The finite computations of the two packages (exact integers and rationals, exact algebraic certificates in $\mathbb{Q}(\sqrt{5})$) check identities, transcriptions and the amplitude digits of Part I; no asymptotic statement rests on them.

The OEIS entries

The write read the four entries on 7 October 2026; the revisions are those the Parts cite. **A094926** [1] (revision #27 of 19 April 2016), by Yasutoshi Kohmoto, records in its formula field “ $a(n) \sim c*\phi^n$ with $\phi=1.61803\dots$ being the golden ratio and $c = A258639 = 0.54172002195814443386932\dots$ (conjectured). - Manfred Scheucher, Jun 03 2015”. **A094925** [2] (revision #20 of 4 May 2021), with offset 1, records “ $a(n) \sim c*\phi^n \dots c =$ ” followed by

0.78529667298898361017570049509486675274402985275383398273772345738007479334754...

“(conjectured). Cf. A094926. - Manfred Scheucher, Jun 03 2015”. **A258639** [3] (revision #4 of 6 June 2015), by Scheucher and Kotěšovec, defines its constant as $\lim_n A094926(n)/\phi^n$. Part I’s Theorem 1.1

proves $a_n \sim C\phi^n$ for both seeds, and its certificates (Theorem 4.1) agree with every printed digit: C_0 with A094926's and all 106 supplied digits of A258639, and C_1/ϕ , the amplitude in A094925's own indexing, with all 77 digits in A094925. So both of Scheucher's conjectures are **proved** as stated, and the limit that A258639 presupposes exists. **A078510** [6] (revision #21 of 11 September 2025), by Neil Fernandez (5 January 2003), records the delayed recurrence through A265409 (Antti Karttunen, 13 December 2015) and no asymptotic formula; Part II's equivalent settles no OEIS conjecture, and Report 191 claims none. Nothing was submitted to the OEIS. The OEIS terms and digits in `data/190-hex-oeis_fixtures.json` and `data/191-square-oeis_fixtures.json` are OEIS content under CC BY-SA 4.0.

[write] **7 October 2026; Scheucher's heuristic form.** A094926 also carries Scheucher's comment of 3 June 2015, which reads in part "I believe that $a(n) = (a(n-1) + a(n-2)) / (1 - c \cdot d^{\sqrt{n}})$ can be proven properly". This heuristic is offered as a route to the equivalent, which Part I proves. Its literal form cannot hold with constants c, d , even asymptotically. Indeed, by (12), $a_n / (a_{n-1} + a_{n-2}) - 1 = \sum_{j \in E_n} a_j / (a_{n-1} + a_{n-2})$; since $a_j = C\phi^j(1 + o(1))$, every $j \in E_n$ tends to infinity with n , E_n has at most two elements, and $\phi^{n-1} + \phi^{n-2} = \phi^n$, this equals $\eta_n(1 + o(1))$ with η_n of (16). By (17), at the final corner of a side $\eta_n = Q^r \phi^{3-t}$, while at the row before it, on the same side and stage, $\eta_{n-1} = Q^r \phi^{5-t}$. So the ratio of consecutive excesses tends to ϕ^{-2} along the corner rows, whereas the form $c d^{-\sqrt{n}} / (1 - c d^{-\sqrt{n}})$ would make it tend to 1, because $\sqrt{n} - \sqrt{n-1} \rightarrow 0$. The excess is $c_n d^{-\sqrt{n}}$ with $d = \phi^{2\sqrt{3}}$ only if c_n is allowed to oscillate with the position on the spiral, as Part I's phase theorem quantifies for a_n itself (Theorem 1.2). This bears only on the comment, not on the conjectured equivalent, which is proved.

The inverses and the transseries volume

Both Parts compare their inversion with the repository's transseries volume [18], Report 190 at revision `6bf7f30d0` and Report 191 at `9ba11eaf7`; both revisions exist, and the passages they name are there. Its symbols are subscripted "vol" here. Statement by statement:

- Part I's Theorem 9.2: its first step, $N(y) = \lceil X(\log y) \rceil$ with X the inverse of the piecewise-linear interpolation L of $\log a_n$ on a late half-line, is an *instance* of item (1) of `p0:thm:staircase`, with $A_n = \log a_n$ (strictly increasing there), the admissible interpolation $\mathcal{A}_{\text{vol}} = L$ and target $\log y$. The bracket (66) then follows from item (1) and the monotonicity of the ceiling, as item (2) does; the estimate (64) that feeds it is Part I's own.
- Part I's Theorem 9.1: proved directly from the sign and size of the deficit, without an interpolation; an *analogue*, which exhibits the failure mode of item (2) for the bare leading chart.
- Part II's (127): an *instance* of item (1) of `p0:thm:staircase`, with $A_n = \log a_n$ for $n \geq 7$ and the linear interpolation $\mathcal{A}$; the estimate of Theorem 12.2 is Part II's own.
- Part II's $V = 2W_0(\sqrt{2L})$ in (132), the root of $V^2 e^V = 8L$, that is of $V + 2 \log V = \log(8L)$: an *instance* of `p0:thm:lambert-core` with $a_{\text{vol}} = 1$, $b_{\text{vol}} = 2$ and $L_{\text{vol}} = \log(8L)$, which gives $X = 2W_0(\frac{1}{2}e^{\log(8L)/2}) = 2W_0(\sqrt{2L})$.
- Part II's germ δ of Lemma 19.1: as a formal series, an *instance* of `p0:thm:lambert-centered` with $R_{\text{vol}} = \mathbb{Q}$, $a_{\text{vol}} = 1$, $b_{\text{vol}} = 2$, $Q_{\text{vol}}(s) = \log(1 - s + s^2)$ and $t = u = 1/V$: putting $w_0 = V(1 + u\delta)$ in $w_0 + 2 \log w_0 + \log(1 - 1/w_0 + 1/w_0^2) = V + 2 \log V$, which is (74) in logarithmic form, gives $\delta = -[2 \log(1 + u\delta) + Q_{\text{vol}}(u/(1 + u\delta))]$, the volume's fixed-point equation $U = \Psi(t, U)$, and this is (133). The coefficients $1, -\frac{5}{2}$ of (137) follow from it. The convergence of the germ, and the hierarchy of Theorem 19.2, are Part II's own.

Neither Part claims novelty for the inversion mechanics. Report 191 also cites *Transseries for Mere Mortals* as a “supplied manuscript”; that tutorial is one of the four sources merged into the volume’s Part Q0 (see the volume’s Section “Why”, label `q0:sec:why`).

Notation across the two Parts

Since the Parts treat different sequences, most letters mean different things in each. No symbol was renamed; each Part fixes its own meanings. The ones a reader is most likely to carry over wrongly:

Symbol	Meanings
a_n, C	Part I: either hexagonal sequence (seed superscript suppressed) and its amplitude $C = \lim a_n \phi^{-n}$; C_1/ϕ is A094925’s amplitude in its own indexing. Part II: A078510 and the amplitude of $e^{H(n)}$. <i>False reading</i> : the two C are unrelated numbers on different scales.
r, t	Part I: $r = r(n)$ the stage of the hexagonal spiral (5); t a side label $0, \dots, 5$. Part II: $r = \sqrt{x}$; $t(n)$ the delayed predecessor (81). <i>False reading</i> : Part II’s t is an index function, not a side.
Q, ρ, α, β	Part I: $Q = \phi^{-6}$, $\rho = -\phi^{-2}$, $\alpha = \phi/\sqrt{5}$, $\beta = 1/(\phi\sqrt{5})$ (1). Part II: $Q(u, \delta)$ in (134), Q_0 ; α_n, β_n convex weights (109).
H, F, D	Part I: H_n the stable filter (36); F_k the correction coefficients; D_m the delay operator (31), D_n the normalized deficit. Part II: $H = \Phi + \Psi$ the normalizer (69); $F = H + \Xi$; D_j memory diameters; $d(n) = n - t(n)$.
θ, Φ, Ψ, G	Part I: $\theta = \{\sqrt{12n}\}$ and $G(\theta)$ the phase factor (7). Part II: $\theta(x) = \{2\sqrt{x}\}$; Φ, Ψ parts of H ; $G(r)$ in Proposition 17.2.
E, η, T	Part I: E_n the extra neighbour set, η_n the forcing weight (16), T_n its tail. Part II: E the exceptional corner indices (93); T_j an error tail.
L, U, N, X, K	Part I: L_r, U_r stage endpoints; L an interpolation of $\log a_n$; $N(y)$ the threshold; X its inverse; K_J a constant. Part II: $L = \log Y$; $N(Y)$ the threshold; $X(L)$; K, K_*, K_J constants.
W, w	Part I: W_r the stage weight sum (18); $w(x)$ a profile weight. Part II: W_0 the Lambert function, $w = W_0(4\sqrt{x})$.

Provenance and merge decisions

Two manuscripts of the session bundle of Reports 1–243, both dated 4 October 2026. Their title blocks read “Report190” and “Report191”, with the subtitles above in the author slot; both PDF author fields read “Research report”; neither names an author, a tool or an addressee, and neither carries “prepared for private review” wording. Report 190 names the ProveIt revision 6bf7f30d0352f7596e70928b3d4f304914075907 (4 October 2026) as the read-only comparison point for the transseries volume, and Report 191 names 9ba11eaf7c1b6a27ccfc0bab63e22f97768cb704 (3 October 2026) for the volume and the Lambert W Guide (`Analysis/FabiusFunction/docs/s-emi-formalized-research-frontiers/drafts/lambert-w/Lambert_W_Guide/`); both commits exist and contain those files. They arrived in commit 60f54ea06 and were placed by 8622ca7e5 (batch 110, cluster 110-SPIRAL).

- **Part I**, bundle Report 190, archive `Hexagonal_Spiral_Fibonacci_Asymptotics_and_Inverses_Source.zip` (789,595 bytes; 22 files; `Report190.tex`, 660 lines, 15 pages); the base.
- **Part II**, bundle Report 191, archive `Square_Spiro_Fibonacci_Growth_and_Inversion_Source.zip` (628,802 bytes; 23 files; `Report191.tex`, 766 lines, 19 pages). It cites Report 190 as `report190`; that bibliography entry now points to Part I.

Where the merge had to choose:

1. *One report or two.* The intake’s dossier offered both: one report in two Parts, or two single reports. The placement chose one report; the write keeps it, and states in this Guide that the Parts share a construction family and a method but no theorem.
2. *Base and order.* Report 190 first (see the Guide).
3. *Front matter, numbering and bibliography.* The two title blocks and abstracts are printed at the heads of their Parts, with each manuscript’s opening scope paragraph; one table of contents replaces two. The two bibliographies are merged; the one shared key, **fernandez**, is one entry keeping both Parts’ details, and the write added the entry for the transseries volume.
4. *Write additions.* This front matter, Section 23, the dated notes and the label prefixes. Everything else is delivered text.

What was checked, and what was not

The report is unrefereed and not formalized. At intake (dossier of batch 110, 6 October 2026) both manuscripts were read and no mathematical error was found; both mandatory exact checkers were rerun on copies, in normal and optimized Python, and reproduced their recorded outputs byte for byte. The write (7 October 2026) read the four OEIS entries and compared every printed OEIS digit with Part I’s certified table; checked the identifications with the transseries volume above, including the first two coefficients of δ ; proved the note on Scheucher’s heuristic form; and reran both exact checkers on copies (results in the README). What was *not* done: the analytic proofs were read and their displayed algebra spot-checked, but they are the manuscripts’ proofs, not an independent verification; Kundrát’s, Appleby–Buckwar’s and Ryde’s texts, Fernandez’s page and the Lambert W Guide were not re-read; the optional floating-point diagnostics and the release builders were not run.

Relation to neighbouring reports and formal projects

No other placed report treats A094926, A094925, A258639, A078510, A265409, A265370, A265404 or spiral-Fibonacci recurrences (repository search, 6–7 October 2026). The transseries volume [18] and the Lambert W Guide are related to the inverses as stated above; Report 191’s use of the Guide is motivational (a constant-delay Lambert balance). Placement in the collection confers no formal status: no Lean or Rocq file in the repository treats these sequences, and no statement of this report is formalized.

Part I

Hexagonal spiral Fibonacci asymptotics: certified amplitudes and all fixed order feedback corrections

Bundle Report 190, archive `Hexagonal_Spiral_Fibonacci_Asymptotics_and_Inverses_Source.zip`, delivered as `Report190.tex` (15 pages), the base of the merge. Delivered title block “Report190 / Hexagonal Spiral Fibonacci Asymptotics / Certified amplitudes and all fixed order feedback corrections / 4 October 2026”. Printed as delivered, with `[write]` notes; section, statement and equation numbers are those of the delivery. The opening paragraph “Scope and attribution” is the manuscript’s own. This Part is self-contained; it uses nothing from Part II.

Abstract of this Part (as delivered). The hexagonal spiral Fibonacci sequences A094926 and A094925 satisfy $a_n \sim C\phi^n$, proving the two leading equivalents conjectured in their OEIS records in 2015. We derive their exact six-side delay recurrence, prove convergence by a positive Perron projection, and give algebraic amplitude bounds that certify 120 truncated fractional digits. After normalization, both seed choices share a universal expansion at every fixed stretched-exponential order. Its coefficients are exact iterates of a geometry-defined operator; the finite corner layers are retained. The leading relative deficit is

$$\sqrt{n} e^{-2\sqrt{3}\log(\phi)\sqrt{n}} \left\{ \frac{\phi^{4+\theta}(\phi^2 - \theta)}{\sqrt{15}} + \mathcal{O}(n^{-1/2}) \right\}, \quad \theta = \{\sqrt{12n}\},$$

with a continuous periodic phase factor and explicit sharp lower and upper envelopes. Every higher sector has a nonzero leading profile. We also give fixed-order real inverse estimates and pointwise ceiling brackets. The uncorrected logarithmic ceiling is too small by one infinitely often, even for integer targets. An exact standard-library verifier and a deterministic offline build accompany the proofs.

Scope and attribution. The sequences, the predicted golden-ratio growth, the square-root delay motivation, and the previously displayed amplitudes are prior work. Binet’s formula, variation of constants, positive product bounds, and monotone interpolation are classical. This report supplies the target-specific proofs, uniform correction hierarchy, phase envelopes, and certificates below. All expansion orders are fixed; no convergent infinite expansion, effective asymptotic onset, globally exact rounded inverse, or worldwide priority is asserted. Source scope is recorded in Section 11.

1 Conventions and main conclusions

Put

$$\phi = \frac{1 + \sqrt{5}}{2}, \quad Q = \phi^{-6}, \quad \rho = -\phi^{-2}, \quad \alpha = \frac{\phi}{\sqrt{5}}, \quad \beta = \frac{1}{\phi\sqrt{5}}. \quad (1)$$

The letter Q is a positive stage ratio; ρ is the negative stable Fibonacci root after normalization. Two sequences use the same geometric recurrence but different central seeds:

$$a_0^{(0)} = 0, \quad a_1^{(0)} = 1; \quad a_0^{(1)} = a_1^{(1)} = 1. \quad (2)$$

Their relation to the original OEIS indices is

$$A094926(n) = a_n^{(0)}, \quad A094925(m) = a_{m-1}^{(1)} \quad (m \geq 1). \quad (3)$$

Accordingly, if $C_\sigma = \lim_n a_n^{(\sigma)} \phi^{-n}$, the amplitude multiplying ϕ^m in A094925 is C_1/ϕ , not C_1 . We suppress the superscript when a statement applies to either seed.

The recurrence is defined precisely in Section 2. Its stage endpoints and index function are

$$L_r = 3r^2 - 2r + 2, \quad U_r = 3r^2 + 4r + 2 \quad (r \geq 1), \quad (4)$$

$$r(n) = \left\lfloor \frac{1 + \sqrt{3n - 5}}{3} \right\rfloor \quad (n \geq 3), \quad r(n) = 0 \quad (0 \leq n \leq 2). \quad (5)$$

Thus $L_r \leq n \leq U_r$ exactly when $r(n) = r$.

Theorem 1.1 (Amplitudes and fixed orders). *For both sequences in (2), a finite constant $C > 0$ exists and $a_n \sim C\phi^n$. Define $F_0(n) = 1$ and $F_{k+1} = SF_k$, where the operator S in (32) is explicit and depends only on the spiral geometry. For every fixed integer $J \geq 0$,*

$$\frac{a_n}{C\phi^n} = \sum_{k=0}^J F_k(n) + \mathcal{O}_J\left((1 + r(n))^{J+1} Q^{(J+1)r(n)}\right). \quad (6)$$

All defining series are absolutely convergent. The functions F_k are the same for both seed choices, are computable to arbitrary prescribed absolute accuracy, and have the nonzero profiles in Theorem 7.1. The theorem is a finite-order asymptotic statement, not a claim of convergence as $J \rightarrow \infty$.

Theorem 1.2 (Sharp square-root phase). *Extend*

$$G(\theta) = \frac{\phi^{4+\theta}(\phi^2 - \theta)}{\sqrt{15}}, \quad 0 \leq \theta \leq 1, \quad (7)$$

continuously and periodically to $\mathbb{R}$. For either normalized seed,

$$1 - \frac{a_n}{C\phi^n} = \sqrt{n} e^{-2\sqrt{3}\log(\phi)\sqrt{n}} \left\{ G(\{\sqrt{12n}\}) + \mathcal{O}(n^{-1/2}) \right\}. \quad (8)$$

The normalized expression

$$D_n = \frac{1 - a_n/(C\phi^n)}{\sqrt{n} e^{-2\sqrt{3}\log(\phi)\sqrt{n}}}$$

has the sharp envelopes

$$\liminf D_n = \frac{\phi^6}{\sqrt{15}}, \quad \limsup D_n = \frac{\phi^{4+\phi^2}}{e\sqrt{15}\log\phi}. \quad (9)$$

In particular, $a_n < C\phi^n$ eventually, and $C\phi^n - a_n \rightarrow +\infty$.

Section 9 proves that the intrinsic integer inverse lies eventually in two adjacent logarithmic-ceiling candidates and that both candidates occur infinitely often. Higher corrections give narrower real-index brackets; they do not remove the need to check an integer rounding boundary. Exact amplitude certificates, unlike those asymptotic inverse estimates, have a fully specified finite stage and no unknown onset.

[write] **7 October 2026.** Theorem 1.1 proves the two equivalents that Manfred Scheucher recorded as conjectures in A094926 and A094925 on 3 June 2015, and Theorem 4.1 agrees with every digit printed there and in A258639 (front matter, “The OEIS entries”, which also shows that Scheucher’s accompanying heuristic form cannot hold with constants).

2 The exact spiral and its delay table

Represent the triangular lattice by the six neighbor vectors

$$D_0 = (-1, -1), \quad D_1 = (-2, 0), \quad D_2 = (-1, 1), \quad D_3 = (1, 1), \quad D_4 = (2, 0), \quad D_5 = (1, -1). \quad (10)$$

The Euclidean regular triangular lattice is obtained by a fixed linear rescaling; adjacency is unchanged. Start with $v_0 = (0, 0)$ and $v_1 = (2, 0)$. Stage $r \geq 1$ takes these directions in order, with lengths

$$\ell_t(r) = r + \mathbf{1}_{\{t=4\}}, \quad 0 \leq t \leq 5. \quad (11)$$

At each new index n , the rule adds a_{n-1} to the values at all earlier neighbors of v_{n-1} . The path predecessor contributes a_{n-2} ; the other earlier neighbor indices form E_n . Thus

$$a_n = a_{n-1} + a_{n-2} + \sum_{j \in E_n} a_j \quad (n \geq 2), \quad E_0 = E_1 = E_2 = \emptyset. \quad (12)$$

This is the adjacency rule in the original records [1, 2].

Proposition 2.1 (Six-side delay formula). *On side t of stage r , the recurrence indices start at $L_r + tr + \mathbf{1}_{\{t=5\}}$, have length $\ell_t(r)$, and have delay parameter*

$$d_{r,t} = 6r - 4 + t. \quad (13)$$

For the first $\ell_t(r) - 1$ indices and for the final index, respectively,

$$E_n = \{n - d_{r,t} - 1, n - d_{r,t}\}, \quad E_n = \{n - d_{r,t} - 1\}. \quad (14)$$

The one-neighbor case at every side endpoint is essential.

Proof. Set $\tau_r = 3r^2 - 2r$ and $s_t(r) = tr + \mathbf{1}_{\{t=5\}}$. Stage r starts at vertex index τ_r and coordinate $(2r, 0)$. The vertex with side position $1 \leq h \leq \ell_t(r)$ has index $\tau_r + s_t(r) + h$. The complete coordinate list is

t	$\ell_t(r)$	Side-start coordinate	Coordinate at position h
0	r	$(2r, 0)$	$(2r - h, -h)$
1	r	$(r, -r)$	$(r - 2h, -r)$
2	r	$(-r, -r)$	$(-r - h, -r + h)$
3	r	$(-2r, 0)$	$(-2r + h, h)$
4	$r + 1$	$(-r, r)$	$(-r + 2h, r)$
5	r	$(r + 2, r)$	$(r + 2 + h, r - h)$

For a displayed vertex, its path predecessor is obtained in direction $-D_t$. The two possible inward neighbors are in directions D_{t+2} and D_{t+1} , subscripts taken modulo six. Substitution in the coordinate list identifies them as preceding-stage side positions $h - 1$ and h , with indices

$$\tau_{r-1} + s_t(r - 1) + h - 1, \quad \tau_{r-1} + s_t(r - 1) + h. \quad (15)$$

The second exists only for $h < \ell_t(r)$: that preceding side is shorter by one. The remaining three direction tests in (10) give either unvisited points of the current path or points outside the occupied spiral. These six substitutions account for every possible earlier neighbor. At $r = 1$, the preceding stage is the degenerate stage zero, with lengths 0, 0, 0, 0, 1, 0, joining v_0 to v_1 ; the same tests apply, including its zero-length sides.

The recurrence row for this vertex is $n = \tau_r + s_t(r) + h + 1$. Since

$$\tau_r - \tau_{r-1} = 6r - 5, \quad s_t(r) - s_t(r-1) = t,$$

subtraction gives (14). The first row is $\tau_r + 2 = L_r$, and the last is $\tau_{r+1} + 1 = U_r$, proving the stage convention as well. $\square$

In particular, each extra delay lies between $6r - 4$ and $6r + 2$. If n belongs to stage r , its extra indices have recurrence stages $r - 1$ or $r - 2$, apart from the harmless initial convention $r(j) = 0$. The geometric inward stage is always the preceding stage; the possible second shift is caused by the two-index offset between vertex stages and recurrence stages. Only a bounded number of side-start or corner positions per stage have this exception. This distinction matters in the profile induction below.

For reference, the first values in the two conventions are

$$\begin{aligned} a_n^{(0)} &: 0, 1, 1, 2, 3, 5, 8, 14, 23, 38, 63, 102, 168, 272, 445, \dots, \\ a_n^{(1)} &: 1, 1, 2, 4, 7, 12, 20, 34, 55, 90, 148, 240, 394, 638, 1043, \dots \end{aligned}$$

The verifier independently walks the coordinates and compares the earlier-neighbor sets with the delay table. Agreement with finitely many rows or listed integer terms supplements the geometric proof; it cannot replace it.

3 Exact forcing weights and geometric tails

Define the normalized geometric forcing weight

$$\eta_n = \sum_{j \in E_n} \phi^{j-n}, \quad \eta_0 = \eta_1 = \eta_2 = 0. \quad (16)$$

Using $1 + \phi^{-1} = \phi$ in (14) gives

$$\eta_n = \begin{cases} Q^r \phi^{5-t}, & \text{at a non-corner on side } t, \\ Q^r \phi^{3-t}, & \text{at its final corner.} \end{cases} \quad (17)$$

Summing the six sides yields

$$W_r := \sum_{n=L_r}^{U_r} \eta_n = Q^r (Ar + B), \quad (18)$$

$$A = 12\phi + 8 = \phi^7(1 - Q), \quad B = -7\phi - 4 = \phi^2 - \phi^6. \quad (19)$$

Indeed, a side contributes $Q^r \{\ell_t(r)\phi^{5-t} - \phi^{4-t}\}$, since $\phi^{5-t} - \phi^{3-t} = \phi^{4-t}$. Summing these six elementary expressions proves (18) in $\mathbb{Q}(\phi)$.

Write $T_n = \sum_{m>n} \eta_m$. The sums of Q^s and sQ^s give the exact complete-stage tail

$$T_{U_R} = Q^{R+1} \left\{ \phi^7(R+1) - \phi^6 + \frac{2\phi}{1-Q} \right\} \quad (R \geq 1). \quad (20)$$

For a row at side t , position h , $1 \leq h \leq \ell_t(r)$, one also has the exact partial-stage formula

$$\begin{aligned} T_n = Q^r & \left\{ (\ell_t(r) - h)\phi^{5-t} - \mathbf{1}_{\{h < \ell_t(r)\}} \phi^{4-t} \right. \\ & \left. + \sum_{u=t+1}^5 (\ell_u(r)\phi^{5-u} - \phi^{4-u}) \right\} + T_{U_r}. \end{aligned} \quad (21)$$

Thus $T_n \in \mathbb{Q}(\phi)$ is exactly computable for every n . In particular,

$$\sum_n \eta_n < \infty, \quad T_n = \mathcal{O}((1 + r(n))Q^{r(n)}). \quad (22)$$

The weight sum, rather than a ratio heuristic, is what controls the amplitude.

4 Proof and exact certification of the amplitudes

Let $b_n = a_n \phi^{-n}$ and $d_n = \sum_{j \in E_n} \phi^{j-n} b_j$. Define the positive Perron projection

$$c_n = \frac{\phi^{1-n}}{\sqrt{5}} (a_n + \phi^{-1} a_{n-1}) \quad (n \geq 1). \quad (23)$$

Direct substitution of (12) gives

$$c_n - c_{n-1} = \alpha d_n \geq 0 \quad (n \geq 2). \quad (24)$$

Also, for $j \geq 1$, $b_j \leq (\sqrt{5}/\phi)c_j$. Every extra endpoint is at least one for $n \geq 10$, so monotonicity gives

$$c_n \leq (1 + \eta_n)c_{n-1} \quad (n \geq 10). \quad (25)$$

The finite prefix is positive, and (22) bounds the subsequent product. Consequently c_n increases to some finite $C > 0$. There is no appeal to an undefined c_0 .

The definition of c_n also yields

$$b_n + \phi^{-2} b_{n-1} = (1 + \phi^{-2})c_n. \quad (26)$$

Subtracting the corresponding identity for the constant C and iterating a stable scalar recurrence proves $b_n \rightarrow C$, since $|\rho| < 1$. This proves the leading equivalents in Theorem 1.1 for both seeds. The same argument works for nonnegative a_0 and positive a_1 .

Theorem 4.1 (Finite algebraic certificates). *For $N = U_R$, $R \geq 1$,*

$$c_N \leq C \leq c_N e^{T_N}. \quad (27)$$

If $T_N < 1$, then the purely algebraic bounds

$$c_N \leq C \leq \frac{c_N}{1 - T_N} \quad (28)$$

are valid. All quantities in (28) belong to $\mathbb{Q}(\phi)$ and are obtained from exact integers and the delay geometry.

Proof. Every future row after $U_R \geq 9$ has extra endpoints at least one. Iterating (25) and using $1 + x \leq e^x$ gives (27). For $0 \leq t < 1$, comparison of the exponential and geometric power series gives $e^t \leq (1 - t)^{-1}$, proving (28). $\square$

These bounds are effective at the specified finite index. Their width tends to zero because $T_N \rightarrow 0$ and $c_N \leq C$. No asymptotic error constant or unknown onset is used in a certificate.

4.1 Rational evaluation and certified digits

For exact arithmetic, represent $u + v\phi$ by $(u, v) \in \mathbb{Q}^2$, use $\phi^2 = \phi + 1$, and invert by its norm

$$(u + v\phi)^{-1} = \frac{u + v - v\phi}{u^2 + uv - v^2}.$$

At $R = 100$, $N = 30402$, take $P = 2N + 512$ and

$$z = \left\lfloor \sqrt{5 \cdot 2^{2P}} \right\rfloor, \quad p_- = \frac{2^P + z}{2^{P+1}}, \quad p_+ = \frac{2^P + z + 1}{2^{P+1}}. \quad (29)$$

Then $p_- < \phi < p_+$. The sign of v chooses the outward endpoint for $u + v\phi$. If $[c_-, c_+]$ and $[t_-, t_+]$ are such rational enclosures for c_N and T_N , the interval

$$\left[c_-, \frac{c_+}{1 - t_+} \right] \quad (30)$$

is a rigorous enclosure for C , provided $0 \leq t_+ < 1$. To enclose C_1/ϕ , divide its lower endpoint by p_+ and its upper endpoint by p_- . The code also recomputes the certificate in the independent basis $u + v\sqrt{5}$.

For readability, the following bounds are rounded outward to 110 fractional places. Spaces split long strings into blocks and carry no numerical meaning. Within each pair the last block alone differs:

Quantity	Outward decimal endpoints, continued over lines
C_0 lower	0.54172002195814443386932280287176743847770577173050013512373 119279413708701041825302581487910167435096508609105
C_0 upper	0.54172002195814443386932280287176743847770577173050013512373 119279413708701041825302581487910167435096508609106
C_1 lower	1.27063670814838696022586698316380373738117229438866227263872 100585311783758622656587107702549959717913615220765
C_1 upper	1.27063670814838696022586698316380373738117229438866227263872 100585311783758622656587107702549959717913615220766
C_1/ϕ lower	0.78529667298898361017570049509486675274402985275383398273772 345738007479334754040183932006451774593289056517183
C_1/ϕ upper	0.78529667298898361017570049509486675274402985275383398273772 345738007479334754040183932006451774593289056517184

The underlying rational intervals are much narrower. For each of C_0 , C_1 , and C_1/ϕ , the verifier checks equality of the two exact integers $\lfloor 10^{120}L \rfloor$ and $\lfloor 10^{120}U \rfloor$, with L, U the rational endpoints, certifying at least 120 truncated fractional digits. The displayed decimal pairs share only 109 fractional digits; they alone do not certify 120. Independently, the exact bounds reproduce all 106 supplied digits of A258639 and all 77 amplitude digits displayed in the A094925 record. Those published digits and approximate amplitudes predate this report; no new-digit priority is claimed.

5 An exact geometry-defined correction operator

For a sequence $f = (f_n)_{n \geq 0}$, define

$$(D_m f) = \sum_{j \in E_m} \phi^{j-m} f_j. \quad (31)$$

For the classes considered below, define

$$(Sf)_n = -\alpha \sum_{\substack{m \geq n \\ m \geq 2}} D_m f + \beta \sum_{m=2}^n \rho^{n-m} D_m f \quad (n \geq 0). \quad (32)$$

The second sum is empty for $n = 0, 1$. The first is an absolutely convergent future tail. This is a specified linear operator involving only E_m , ϕ , and its two root weights; neither the target sequence nor its unknown amplitude enters its definition.

Lemma 5.1 (Exact variation of constants). *Let $b_n = a_n \phi^{-n}$. Then*

$$b = C\mathbf{1} + Sb + B_0(\rho^n)_{n \geq 0}, \quad B_0 = \frac{\phi a_0 - a_1}{\sqrt{5}}, \quad (33)$$

and

$$C = A_0 + \alpha \sum_{m=2}^{\infty} d_m, \quad A_0 = \frac{a_1 + \phi^{-1} a_0}{\sqrt{5}}, \quad d_m = D_m b. \quad (34)$$

Proof. Let $e_m = \sum_{j \in E_m} a_j$. Ordinary Fibonacci numbers satisfy $\text{Fib}_0 = 0$, $\text{Fib}_1 = 1$, and their Binet formula gives the exact recurrence solution

$$a_n = A_0 \phi^n + B_0 (-\phi^{-1})^n + \sum_{m=2}^n \text{Fib}_{n-m+1} e_m. \quad (35)$$

After division by ϕ^n , the last term equals

$$\alpha \sum_{m=2}^n d_m + \beta \sum_{m=2}^n \rho^{n-m} d_m.$$

The plus sign on the second sum follows from the negative second Fibonacci root. The already proved boundedness of b , together with $|d_m| \leq \|b\|_{\infty} \eta_m$, shows absolute summability. The finite stable convolution tends to zero. Taking limits proves (34); splitting its dominant-root sum at n then proves (33), including $n = 0, 1$. $\square$

The first exact coefficient has a particularly short formula. Put

$$H_0 = 0, \quad H_n = \rho H_{n-1} + \eta_n \quad (n \geq 1). \quad (36)$$

Then

$$F_1(n) = S\mathbf{1}(n) = -\alpha T_n + \beta H_n. \quad (37)$$

Both terms are exactly computable in $\mathbb{Q}(\phi)$, using (21). The stable filter H_n keeps the alternating side and corner transition layers. Dropping it, or keeping only a leading stage profile, is generally incompatible with a claimed error as small as the second entire stretched-exponential sector.

6 Mapping bounds and the proof at every fixed order

For a fixed integer $k \geq 0$, let X_k be the space of real sequences with finite weighted norm

$$\|f\|_k = \sup_{n \geq 0} \frac{|f_n|}{(1 + r(n))^k Q^{kr(n)}}. \quad (38)$$

The constants in all ensuing bounds may depend on the fixed order, but not on the index or phase.

Lemma 6.1 (One-order improvement). *The operator S maps X_k boundedly into X_{k+1} . Its stable backward part alone satisfies the sharper estimate*

$$\sum_{m=2}^n \rho^{n-m} D_m f = \mathcal{O}_k \left(\|f\|_k (1 + r(n))^k Q^{(k+1)r(n)} \right). \quad (39)$$

Proof. If m is in stage $s \geq 1$, its extra indices have stages between $\max(0, s-2)$ and s . From (17),

$$|D_m f| \leq A_k \|f\|_k (1+s)^k Q^{(k+1)s}, \quad A_k = \phi^5 Q^{-2k}. \quad (40)$$

There are $6s+1 \leq 7(1+s)$ rows in stage s . For $r = r(n)$, a tail therefore has absolute value at most

$$7A_k \|f\|_k \sum_{s \geq r} (1+s)^{k+1} Q^{(k+1)s} \leq 7A_k Z_k \|f\|_k (1+r)^{k+1} Q^{(k+1)r}, \quad (41)$$

where $Z_k = \sum_{h \geq 0} (1+h)^{k+1} Q^{(k+1)h} < \infty$.

For the backward sum, $m \leq n$ implies $s = r(m) \leq r = r(n)$. Moreover

$$r - s \leq \sqrt{n - m}. \quad (42)$$

To check this, when $t = r - s \geq 1$ the smallest possible difference is at least

$$L_r - U_s = 3t^2 - 2t + 6s(t-1) \geq t^2.$$

The same inequality holds for $s = 0$, $m \leq 2$, using $U_0 = 2$; the case $r = s$ is immediate. Hence, with $h = n - m$, (40) bounds the absolute backward sum by

$$A_k \|f\|_k (1+r)^k Q^{(k+1)r} \sum_{h \geq 0} |\rho|^h Q^{-(k+1)\sqrt{h}}. \quad (43)$$

The series converges: its logarithmic summand is a negative linear function of h plus a positive multiple of $\sqrt{h}$. This proves (39), absolute convergence, and the full mapping statement. $\square$

Proof of Theorem 1.1. The amplitude assertion was proved in Section 4. Divide (33) by C and substitute it into itself finitely many times. For every fixed $J \geq 0$, the exact remainder identity is

$$\frac{b}{C} = \sum_{k=0}^J F_k + S^{J+1} \left(\frac{b}{C} \right) + \frac{B_0}{C} \sum_{k=0}^J S^k (\rho^n)_{n \geq 0}. \quad (44)$$

Since b is bounded, the first remainder belongs to X_{J+1} by Lemma 6.1. The sequence (ρ^n) belongs to every X_M , because $n = 3r(n)^2 + \mathcal{O}(r(n))$. Each of its fixed S -iterates also belongs to every X_M , by applying the mapping lemma from a sufficiently strong input space and using the nesting of these spaces. Thus the entire transient remainder is beyond every fixed order. Equation (6) follows. Only finite substitution was used. $\square$

Corollary 6.2 (Seed independence beyond every fixed order). *For every fixed integer $M \geq 0$,*

$$\frac{a_n^{(0)}/C_0}{a_n^{(1)}/C_1} = 1 + \mathcal{O}_M \left((1 + r(n))^M Q^{Mr(n)} \right). \quad (45)$$

Proof. Subtract the normalized expansions to a sufficiently high fixed order and use that both normalized sequences tend to one. This is asymptotic indistinguishability at the stated scales, not an equality of the two sequences. $\square$

6.1 Constructive meaning and truncation scope

The mapping proof provides explicit finite norm bounds, so the word “computable” in Theorem 1.1 need not be interpreted formally. For example replace the series in (43) by the larger algebraic bound

$$V_k^* = \sum_{h=0}^{M_k-1} |\rho|^h Q^{-(k+1)\lceil\sqrt{h}\rceil} + \frac{\phi^{-M_k}}{1-\phi^{-1}}, \quad M_k = 144(k+1)^2. \quad (46)$$

For $h \geq M_k$, the inequality $6(k+1)\lceil\sqrt{h}\rceil \leq h$ implies $|\rho|^h Q^{-(k+1)\lceil\sqrt{h}\rceil} \leq \phi^{-h}$, proving the bound. The sum Z_k in (41) is rational in Q^{k+1} , as follows by repeated application of $q d/dq$ to $(1-q)^{-1}$. Therefore

$$\|Sf\|_{k+1} \leq A_k(7\alpha Z_k + \beta V_k^*)\|f\|_k \quad (47)$$

is a fully specified algebraic norm estimate. Starting from $\|F_0\|_0 = 1$ yields explicit bounds for every fixed F_k .

For fixed $n \leq U_R$, truncating the future sum defining $SF_k(n)$ after U_R loses at most

$$\alpha A_k \|F_k\|_k \sum_{s>R} (6s+1)(1+s)^k Q^{(k+1)s}. \quad (48)$$

This is a computable polynomial-geometric tail tending to zero. The remaining sum has finitely many arguments of F_k ; recursive evaluation with outward rational arithmetic and prescribed error budgets terminates at $F_0 = 1$. Bounds (47)–(48) justify such an interval algorithm for each fixed order and argument. They make no practical complexity claim.

The shipped mandatory checker certifies the amplitudes and finite identities, including exact F_1 identities. Its optional higher-order numerical diagnostic uses a finite cutoff and high-precision floating arithmetic; it does not implement this general interval algorithm or certify the asymptotic inverse constants. Those distinctions keep exact certificates, theoretical computability, and floating diagnostics separate.

7 Nonzero profiles in every correction sector

For n in stage r , define

$$x_n = \frac{n - L_r}{r} \in [0, 6]. \quad (49)$$

The extra vertex on side four changes an actual side coordinate by only $\mathcal{O}(1/r)$. Define a continuous positive piecewise-linear function P on $[0, 6]$ by

$$P(t+\theta) = (\phi^2 - \theta)\phi^{5-t}, \quad 0 \leq t \leq 5, \quad 0 \leq \theta \leq 1. \quad (50)$$

Adjacent formulas agree because $\phi^2 - 1 = \phi$. Write $w(x) = \phi^{5-t}$ for $t < x < t+1$. Then

$$P' = -w \text{ off the side boundaries,} \quad P(0) = \phi^7, \quad P(6) = \phi = QP(0). \quad (51)$$

Theorem 7.1 (Leading profile of each exact coefficient). *For every fixed $k \geq 1$, uniformly over the whole stage,*

$$F_k(n) = \gamma_k r^k Q^{kr} P(x_n)^k + \mathcal{O}_k(r^{k-1} Q^{kr}), \quad \gamma_k = \frac{(-\alpha)^k Q^{-k(k-1)/2}}{k!}. \quad (52)$$

In particular the coefficients have alternating, nonzero leading signs and $F_{k+1}(n)/F_k(n) = \mathcal{O}_k(rQ^r) \rightarrow 0$ uniformly in stage phase.

Proof. For $k = 0$, the starting profile is exactly one. Suppose the profile statement holds at order k . Except for a bounded number of side-start and corner rows in a stage s , each inward recurrence index has stage $s - 1$ and phase $x_m + \mathcal{O}(1/s)$. The profiles P^k are uniformly Lipschitz on $[0, 6]$. For these ordinary non-corner rows, (17) and the induction hypothesis therefore give

$$D_m F_k = \gamma_k Q^{-k} s^k Q^{(k+1)s} w(x_m) P(x_m)^k + \mathcal{O}_k(s^{k-1} Q^{(k+1)s}). \quad (53)$$

At exceptional rows, the inward recurrence stage may be $s - 2$; the corner also has a different weight. Their individual contribution is bounded by $\mathcal{O}_k(s^k Q^{(k+1)s})$. There are only $\mathcal{O}(1)$ such rows per stage. They thus affect the entire or partial stage sum at one lower power of s than its leading s^{k+1} size. They cannot be ignored in an exact coefficient, but do not change its leading profile.

Uniform Riemann summation over a partial stage, allowing its finitely many boundaries, approximates wP^k with error one power of s smaller. Summing complete future stages has geometric ratio Q^{k+1} . Replacing $(r + h)^{k+1}$ there by r^{k+1} has total error $\mathcal{O}_k(r^k)$ after geometric weighting, since

$$|(r + h)^{k+1} - r^{k+1}| \leq C_k r^k (1 + h)^{k+1} \quad (r \geq 1, h \geq 0).$$

Lemma 6.1 makes the stable backward filter one power of r smaller too. The leading new profile is consequently

$$-\alpha \gamma_k Q^{-k} \left\{ \int_x^6 w(u) P(u)^k du + \frac{Q^{k+1}}{1 - Q^{k+1}} \int_0^6 w(u) P(u)^k du \right\}. \quad (54)$$

Since $P' = -w$ and $P(6) = QP(0)$, the expression in braces is exactly $P(x)^{k+1}/(k+1)$. Hence

$$\gamma_{k+1} = -\frac{\alpha Q^{-k}}{k+1} \gamma_k,$$

which gives (52). Positivity and a positive minimum of P make the relative leading-profile error $\mathcal{O}_k(1/r)$ uniformly. This proves the nonvanishing and ratio assertions. $\square$

The first three profile coefficients are

$$\gamma_1 = -\alpha, \quad \gamma_2 = \frac{\alpha^2}{2Q}, \quad \gamma_3 = -\frac{\alpha^3}{6Q^3}.$$

The functions in Theorem 1.1 must remain the exact $F_k = S^k \mathbf{1}$. Replacing F_k by only the leading term in (52) discards $\mathcal{O}(r^{k-1} Q^{kr})$, which is larger than the next whole sector $r^{k+1} Q^{(k+1)r}$. Thus leading profiles describe the hierarchy but do not themselves form the claimed all-fixed-order expansion. The factors $Q^{-k(k-1)/2}$ also underscore why the theorem provides no uniform control in growing k .

8 The square-root phase and its exact envelopes

The fixed-order theorem with $J = 1$, together with Theorem 7.1, gives

$$1 - \frac{b_n}{C} = \alpha r Q^r P(x_n) + \mathcal{O}(Q^r). \quad (55)$$

The next-sector error $\mathcal{O}(r^2 Q^{2r})$ is absorbed here into $\mathcal{O}(Q^r)$. From the exact definition of $x = x_n$,

$$n = 3r^2 + (x - 2)r + 2, \quad (56)$$

so uniformly for $0 \leq x \leq 6$,

$$\sqrt{n/3} = r + \frac{x-2}{6} + \mathcal{O}(1/r), \quad \sqrt{12n} = 6r + x - 2 + \mathcal{O}(1/r). \quad (57)$$

Combining (55) with these relations yields

$$D_n = \frac{\alpha}{\sqrt{3}} \phi^{x_n-2} P(x_n) + \mathcal{O}(1/r). \quad (58)$$

If $x = t + \theta$, (50) collapses the side label and gives precisely $G(\theta)$ from (7). Its endpoint values agree:

$$G(0) = \frac{\phi^6}{\sqrt{15}} = G(1).$$

The periodic extension is globally Lipschitz, so the $\mathcal{O}(1/r)$ phase perturbation in (57) changes G by only $\mathcal{O}(1/r)$ even at a fractional-part wrap. This proves the uniform formula (8).

To find the extrema, differentiate on $(0, 1)$:

$$\frac{d}{d\theta} \log G(\theta) = \log \phi - \frac{1}{\phi^2 - \theta}. \quad (59)$$

It decreases strictly and vanishes at

$$\theta_* = \phi^2 - \frac{1}{\log \phi} \in (0, 1). \quad (60)$$

The interval assertion follows without decimal estimates from $\phi^{-2} < \log \phi < \phi^{-1}$, which is the usual inequality $(z-1)/z < \log z < z-1$ at $z = \phi$. Thus G has its unique maximum at θ_* and its equal minima at the endpoints. Substitution gives (9). Every phase is approached along the within-side grids of mesh $1/r$ as $r \rightarrow \infty$, so these are attained subsequential envelopes, rather than merely pointwise bounds. In fact the whole interval between them is the set of subsequential limits of D_n .

Since G is bounded below by a positive constant, the relative deficit is eventually positive. Multiplication by $C\phi^n$ gives

$$C\phi^n - a_n = \Theta\left(\sqrt{n} \exp\{n \log \phi - 2\sqrt{3} \log \phi \sqrt{n}\}\right) \rightarrow +\infty.$$

The relative error nevertheless tends to zero faster than every inverse power of n . This finishes the proof of Theorem 1.2.

For A094925 in the original index m , replace n by $m-1$ and C by its zero-based value C_1 . Equivalently, the same leading phase formula and envelope constants apply with m and the original-index amplitude C_1/ϕ : the unit shift changes the phase and the scaled prefactor by $\mathcal{O}(m^{-1/2})$. For higher-order accuracy, retain the exact shift $F_k(m-1)$ rather than replacing it by $F_k(m)$.

9 Inverse brackets and the rounding obstruction

For the chosen seed, define the intrinsic inverse

$$N(y) = \min\{n \geq 0 : a_n \geq y\}, \quad x = \log_\phi(y/C). \quad (61)$$

The recurrence is strictly increasing after its finite initial repeated terms, and $a_n \rightarrow \infty$, so this is well defined for positive y .

Theorem 9.1 (Two candidates and infinitely many failures). *For all sufficiently large y ,*

$$N(y) \in \{\lceil x \rceil, \lceil x \rceil + 1\}. \quad (62)$$

Each alternative occurs infinitely often even when y is required to be an integer.

Proof. The eventual strict deficit $a_n < C\phi^n$ excludes every sufficiently large $n < x$ as a crossing index; small indices are excluded when y is large. Hence $N(y) \geq \lceil x \rceil$. Uniformly in the fractional part of x ,

$$\frac{a_{\lceil x \rceil + 1}}{y} = \phi^{\lceil x \rceil + 1 - x}(1 + o(1)) > 1$$

eventually, proving the upper bound.

For integer $y = a_n$, convergence and the negative relative correction imply $n - 1 < x < n$ eventually, so $\lceil x \rceil = n = N(y)$. For $y = a_n + 1$, the divergence $C\phi^n - a_n \rightarrow \infty$ implies again $n - 1 < x < n$ eventually. But strict eventual increase and integrality give $N(a_n + 1) = n + 1$. The uncorrected ceiling therefore misses by exactly one infinitely often. $\square$

9.1 Real interpolation that preserves the fixed-order precision

For fixed $J \geq 0$, set

$$U_J(n) = \sum_{k=0}^J F_k(n), \quad f_J(n) = C\phi^n U_J(n). \quad (63)$$

These values are eventually positive and $f_J(n+1)/f_J(n) \rightarrow \phi$. Let L be the piecewise-linear interpolation of $\log a_n$, and let L_J be the piecewise-linear interpolation of $\log f_J(n)$, both restricted to a sufficiently late half-line. Their slopes tend to $\log \phi$, and hence are eventually at least $(\log \phi)/2$. Denote their increasing inverses by $X = L^{-1}$ and $X_J = L_J^{-1}$.

Theorem 9.2 (All fixed-order inverse envelopes). *Extend $r(t) = \lfloor (1 + \sqrt{3t - 5})/3 \rfloor$ to real $t \geq 3$. For each fixed $J \geq 0$,*

$$|X(\log y) - X_J(\log y)| = \mathcal{O}_J\left((1 + r(x))^{J+1} Q^{(J+1)r(x)}\right). \quad (64)$$

Consequently, for some sufficiently large fixed $K_J > 0$ and all sufficiently large y , put

$$\delta_J(y) = K_J(1 + r(x))^{J+1} Q^{(J+1)r(x)}. \quad (65)$$

Then the pointwise integer bounds are

$$\lceil X_J(\log y) - \delta_J(y) \rceil \leq N(y) \leq \lceil X_J(\log y) + \delta_J(y) \rceil. \quad (66)$$

Their real bracket width tends to zero; the integer endpoints are equal or adjacent eventually.

Proof. Since $U_J(n) \rightarrow 1$, the logarithm is uniformly Lipschitz on the eventual range of both normalized approximations. Theorem 1.1 therefore gives

$$|\log a_n - \log f_J(n)| = \mathcal{O}_J((1 + r(n))^{J+1} Q^{(J+1)r(n)}).$$

Linear interpolation preserves this bound between integers: adjacent stage weights differ by at most a constant depending only on J , even at a stage boundary. Also $L(t) = \log C + t \log \phi + o(1)$ and the same holds for $L_J(t)$. Thus

$$X(\log y) = x + o(1), \quad X_J(\log y) = x + o(1).$$

The stage weights near these two roots and at x are comparable by a fixed factor. At one root, the vertical discrepancy between L and L_J has the stated order; dividing by the lower slope bound for the other interpolation controls the horizontal discrepancy. This proves (64) without assuming differentiability across an integer.

On the eventual increasing half-line, $N(y) = \lceil X(\log y) \rceil$ exactly. Monotonicity of the ceiling gives (66). Its real width $2\delta_J(y)$ is eventually less than one, which proves the final assertion. $\square$

The constants K_J and the onsets in this theorem are asymptotic existence statements, not numerical interval certificates supplied by the package. If an independently justified numerical bracket stays within one ceiling cell, the common endpoint determines $N(y)$ exactly. Without such separation, an unconditional rounded-root formula is unwarranted, however small the real error. This is an application of classical monotone interpolation and prior staircase/error-transport methodology [4]; it is not a new general inversion principle. For the original index of A094925, add one to the zero-based inverse throughout.

[write] 7 October 2026. In the transseries volume’s terms (front matter, “The inverses and the transseries volume”), the identity $N(y) = \lceil X(\log y) \rceil$ used in the proof is an instance of item (1) of `p0:thm:staircase`, with the piecewise-linear L as admissible interpolation, and (66) follows from it as item (2) does. Theorem 9.1 is a direct argument, an analogue.

10 Verification and reproducible deliverables

The public package separates mandatory exact checks from optional numerical diagnostics. Its exact computations use only Python integers and `fractions.Fraction`; no third-party Python module, network, or TeX installation is required to run them. The core commands are

```
python3 -I -S -B code/check_exact.py
python3 -I -S -B -O code/check_exact.py
```

The normal and optimized modes must produce byte-identical canonical JSON. Explicit exceptions implement the input and consistency guards; correctness checks do not depend on Python assertions that optimization could remove. The replay script runs both modes in isolated environments and verifies their agreement.

The checks include independent coordinate and delay-table generation through 1162 recurrence rows, the 79 listed integer terms across the two OEIS records, exact stage-weight identities, and the stage-100 amplitude certificates. A separately implemented $\mathbb{Q}(\sqrt{5})$ arithmetic route cross-checks the main $\mathbb{Q}(\phi)$ arithmetic. Fixture integrity, type and domain errors, and relevant corruption cases are rejected. The receipt states each actually executed finite test and its scope. General asymptotic statements remain theorems of this manuscript, not deductions from a finite test count.

The optional `code/diagnose_float.py` evaluates exact-coefficient formulas with finite-tail truncation using `mpmath`. Higher-order residuals at a sample late stage illustrate the alternating hierarchy. They are noninterval diagnostics, with a specified precision and cutoff, and neither certify digits nor prove remainder constants. The mandatory build never runs them.

To reproduce the exact results and build the complete release from a source tree or intact extracted archive, use the documented commands in `README_REPRODUCIBILITY.md`. The builder creates fresh outputs, runs normal and optimized tests, compiles three isolated pdfLaTeX passes with shell escape disabled, and rejects final warnings, unresolved references, missing glyphs, and overfull or underfull boxes. The source inventory and generated manifest are closed and hash-checked. Archive entry order, timestamps, permissions, and storage mode are fixed. With the same installed

Python/TeX stack, two fresh extracted builds must reproduce the PDF, TeX, ZIP, and external receipt byte for byte. Cross-version identity is not promised.

[write] 7 October 2026. In this report the package’s programs are `code/190-hex-*.py` (those of its `code/` directory and its root scripts), its fixtures and recorded outputs `data/190-hex-*`, and its `README_READUCIBILITY.md`, `DATA_SOURCES.md`, `code/README.md` and `data/README.md` are `190-hex-*.md`. The delivered PDF, the package `README.md` (replaced by this report’s) and `SHA256SUMS.json` are not shipped; the replay and build scripts check the closed delivered inventory, so they run only in a re-extracted archive. The report’s `README` gives a route that runs the mandatory checker on a copy.

The archive contains the authored article and code, bounded attributed integer and digit fixtures, and generated verification products. It excludes full OEIS exports, third-party full text, private research or audit material, transient logs, and unrelated files. SHA-256 manifests detect modification; they are not cryptographic publisher signatures. Reproduction is an offline tool for this authored package, not a sandbox for arbitrary hostile TeX or Python.

11 Sources and the limits of the comparison

Yasutoshi Kohmoto’s OEIS records define the two hexagonal adjacency sequences [1, 2]. The explicit leading equivalents and the square-root-delay motivation are credited there to Manfred Scheucher on 3 June 2015. A258639 credits its amplitude digits to Scheucher and Václav Kotěšovec [3]. The source fixtures in this package are bounded excerpts: 41 and 38 integer terms, 106 amplitude digits, and the separate 77-digit A094925 amplitude string. They are attributed and checked exactly. The source records inspected were revisions 27, 20, and 4, respectively. The linked Sage scripts and full b-files were not accessible in the source review, so no claim is made to have inspected or verified their entire contents.

Fernandez’s *Spiro-Fibonacci Sequences* [5], last modified 11 January 2003, develops neighboring spiral constructions, including the square-spiral model and seed linearity. That construction differs from the present hexagonal rule, which adds all earlier neighbors of the preceding vertex. The actual text and the neighboring OEIS records A078510, A079421, and A079422 were considered; they supply context, not the hexagonal all-order theorem proved here.

The ProveIt transseries and inversion material [4] already treats real-argument Fibonacci continuations, exponential-oscillatory inversion, and staircase/error transport. Those general inverse methods are antecedents. Here the model has geometric delays of order $\sqrt{n}$, stage-scale corrections $e^{-2\sqrt{3}\log(\phi)\sqrt{n}}$, and discrete corner layers. The present target-specific contribution is the exact delay reduction, the amplitude proof and certificates, the fixed-order operator hierarchy with nonzero profiles, and the sharp periodic phase. A bounded search of the selected identifiers, amplitude constants, and spiral-growth terminology found no directly matching proof in the sources inspected. This is a statement of search scope, not exhaustive clearance or a claim of worldwide novelty.

No closed elementary expression for either amplitude is claimed. No convergence in growing correction order, uniform seed-parameter theorem, effective inverse onset, or universal exact rounding rule is asserted. All external source activity was read-only; the report does not imply an OEIS submission or other publication.

[write] 7 October 2026. The square-spiral model and A078510, named above as context, are the subject of Part II, which proves its own growth theorem by a different argument; neither Part uses the other. The write read the live OEIS entries on 7 October 2026; the revisions cited above are current. Nothing was submitted to the OEIS.

Part II

Square Spiro-Fibonacci growth and inversion: positive amplitude and quantitative asymptotics for A078510

Bundle Report 191, archive `Square_Spiro_Fibonacci_Growth_and_Inversion_Source.zip`, delivered as `Report191.tex` (19 pages). Delivered title block “Report191 / Square Spiro Fibonacci Growth and Inversion / Positive amplitude and quantitative asymptotics for A078510 / 4 October 2026”. Printed as delivered, with [write] notes; Report 191’s Section k is Section $k + 11$ here and its statements move with their sections; its equation (k) is equation $(k + 66)$ here. The opening paragraph “Attribution and scope” is the manuscript’s own. This Part is self-contained; it uses nothing from Part I.

Abstract of this Part (as delivered). The square Spiro-Fibonacci sequence A078510 adds the values at the two closest earlier cell centers along a square spiral. Its exact recurrence has a near-identity delayed index $t(n) = n - 4\sqrt{n} + \mathcal{O}(1)$. We prove the full equivalent

$$a_n = C \frac{\exp\left\{\frac{\sqrt{n}}{2}\left(w - 1 + \frac{1}{w}\right) + \frac{w^2}{32} + \frac{w}{2}\right\}}{\sqrt{1+w}} \left(1 + \mathcal{O}\left(\frac{w^3}{\sqrt{n}}\right)\right), \quad w = W_0(4\sqrt{n}), \quad C > 0.$$

An occupied-rectangle induction identifies the exact geometry. A periodic Bernoulli normalizer cancels a nonsummable local defect away from a sparse corner set. Positive products first bound the normalized sequence; a separate common-anchor contraction proves convergence and the error above. We quantify the divergent absolute defect of the uncorrected smooth normalizer and the homogeneous contraction product. Threshold inversion retains $\log C$ with absolute error $\mathcal{O}(w^2)$; a separate analytic implicit-function argument yields every fixed inverse-logarithmic order, with an exact rational coefficient generator. No certified digits of C , sharp relative phase correction, or all-orders relative expansion of the sequence are asserted.

Attribution and scope. The spiral sequence, its seed convention, the quarter-square predecessor sequence A265409, and corner repetition are prior constructions [5, 6, 7, 10]. Lambert balance, WKB normalization, positive comparison, implicit series, and inverse-error transport are established methods. The subject here is their detailed implementation for this specific recurrence, including the convergence mechanism and uniform error. The bounded literature comparison in Section 21 is not a worldwide priority claim.

12 Main conclusions and conventions

All logarithms are natural, and W_0 is the positive real Lambert function: $W_0(z)e^{W_0(z)} = z$ for $z > 0$. For a real variable $x > 0$, put

$$r = \sqrt{x}, \quad w = W_0(4r), \quad (67)$$

$$\Phi(x) = \frac{r}{2} \left(w - 1 + \frac{1}{w}\right), \quad \Psi(x) = \frac{w^2}{32} + \frac{w}{2} - \frac{1}{2} \log(1+w), \quad (68)$$

$$H(x) = \Phi(x) + \Psi(x). \quad (69)$$

At integer arguments, subscripts on r_n, w_n may be omitted. Constants in $\mathcal{O}$ bounds are independent of the integer index, including its position relative to a corner. Every asymptotic statement is at infinity. No effective starting index is supplied.

The sequence starts with $a_0 = 0, a_1 = 1$. With distances measured between centers of unit-square cells, every subsequent cell receives the sum of the values at the two closest earlier centers. One is always its path predecessor. The other has index $t(n)$, so

$$a_n = a_{n-1} + a_{t(n)} \quad (n \geq 2). \quad (70)$$

The prescribed seed a_1 is not an application of a two-predecessor rule. The convention $t(1) = 0$ is only bookkeeping. The first terms are

$$0, 1, 1, 1, 1, 1, 1, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 18.$$

The exact formula for $t(n)$ is proved in Section 13.

Theorem 12.1 (Full growth equivalent). *There is a finite strictly positive constant C such that*

$$a_n = Ce^{H(n)} \left(1 + \mathcal{O}\left(\frac{w_n^3}{r_n}\right) \right). \quad (71)$$

In particular,

$$\log a_n \sim \frac{1}{4} \sqrt{n} \log n. \quad (72)$$

The error in (71) is uniform over all integers, not only the interiors of long sides.

The existence of C is the essential additional issue beyond finding a plausible exponent. A formal local balance identifies $w/(4r)$ as a logarithmic slope, but does not imply that any quotient has a limit. Likewise, summable residuals and positivity alone supply bounds in our proof, not convergence. Section 16 supplies the missing contraction argument.

For a real target $Y > 1$, define the integer threshold

$$N(Y) = \min\{n \geq 0 : a_n \geq Y\}, \quad L = \log Y. \quad (73)$$

Let $w_0 > 0$ be the large-target solution of

$$\frac{e^{w_0}}{8}(w_0^2 - w_0 + 1) = L, \quad r_0 = \frac{w_0 e^{w_0}}{4}. \quad (74)$$

Theorem 12.2 (Quantitative threshold inverse). *As $Y \rightarrow \infty$,*

$$N(Y) = r_0^2 - \frac{4r_0}{w_0} \left(\frac{w_0^2}{32} + \frac{w_0}{2} - \frac{1}{2} \log(1 + w_0) + \log C \right) + \mathcal{O}(w_0^2). \quad (75)$$

Consequently,

$$N(Y) \sim \frac{4(\log Y)^2}{(\log \log Y)^2}. \quad (76)$$

In addition, the threshold has the fixed-order inverse-logarithmic hierarchy stated in Theorem 19.2. That hierarchy is distinct from an all-orders relative expansion of a_n .

The formula retains the amplitude even though its digits are not certified here. Dropping $\log C$ omits the term $-(4r_0/w_0) \log C$, which, unless $C = 1$, is much larger than the stated $\mathcal{O}(w_0^2)$ error. An asymptotic inverse with this error cannot in general be rounded once to obtain an exact integer threshold.

13 The square spiral and the exact predecessor

We use the counterclockwise orientation

$$\begin{aligned} P_0 &= (0, 0), & P_1 &= (1, 0), & P_2 &= (1, 1), & P_3 &= (0, 1), \\ P_4 &= (-1, 1), & P_5 &= (-1, 0), & P_6 &= (-1, -1), & P_7 &= (0, -1), \end{aligned}$$

followed by $P_8 = (1, -1)$. Runs have directions east, north, west, south, repeating, with lengths 1, 1, 2, 2, 3, 3, ... Reflection yields the clockwise orientation without changing any distances or selected indices.

Set

$$q_k = \lfloor k^2/4 \rfloor, \quad A_k = \lfloor k/2 \rfloor, \quad \ell_k = \lfloor (k+1)/2 \rfloor \quad (k \geq 2). \quad (77)$$

The corner $C_k = P_{q_k}$ begins a side of length $q_{k+1} - q_k = \ell_k$. Its outgoing unit vector is v_k , and its incoming unit vector is $z_k = v_{k-1}$. Thus $v_{k+1} = -z_k$. Directly from the path, if $k = 4m + s$,

s	C_k	v_k
0	$(-m, m)$	$(0, -1)$
1	$(-m, -m)$	$(1, 0)$
2	$(m+1, -m)$	$(0, 1)$
3	$(m+1, m+1)$	$(-1, 0)$

In particular,

$$P_{q_k+j} = C_k + jv_k \quad (0 \leq j < \ell_k), \quad C_{k+1} = C_k + \ell_k v_k. \quad (78)$$

For integer $n \geq 1$, the unique side with $q_k \leq n < q_{k+1}$ is selected by

$$k = k(n) = \left\lfloor \sqrt{4n+1} \right\rfloor. \quad (79)$$

Indeed $q_h \leq n$ is equivalent to $h^2 \leq 4n+1$; the equivalence follows by considering even and odd h separately.

Lemma 13.1 (Complete occupied rectangle). *Use local coordinates (a, b) in the orthonormal basis (z_k, v_k) with origin C_k . Immediately before index q_k is placed, the set of all earlier centers is exactly*

$$\mathcal{Q}_k = \{(a, b) \in \mathbb{Z}^2 : -A_k \leq a \leq -1, 0 \leq b \leq \ell_k - 1\}. \quad (80)$$

Immediately before index $q_k + j$, where $0 \leq j < \ell_k$, the earlier set is $\mathcal{Q}_k \cup \{(0, h) : 0 \leq h < j\}$.

Proof. For $k = 2$, $C_2 = (1, 0)$, $z_2 = (1, 0)$, and $v_2 = (0, 1)$. The sole earlier center P_0 has local coordinate $(-1, 0)$, as required. Adding all ℓ_k side cells fills the enlarged rectangle

$$-A_k \leq a \leq 0, \quad 0 \leq b \leq \ell_k - 1.$$

Relative to the next corner C_{k+1} and basis $(v_k, -z_k)$, its coordinates become

$$a' = b - \ell_k, \quad b' = -a.$$

They range over $-\ell_k \leq a' \leq -1$ and $0 \leq b' \leq A_k$. Since $A_{k+1} = \ell_k$ and $\ell_{k+1} = A_k + 1$, this is exactly (80) at $k+1$. The within-side description follows by adding the already placed side points. In particular this is an equality of occupied sets, not only a bounding rectangle. Its cardinality $A_k \ell_k = q_k$ agrees with the number of earlier indices. $\square$

Proposition 13.2 (Exact delayed index). *For every $n \geq 2$, with $k = k(n)$,*

$$t(n) = n - 2k + 3 + \mathbf{1}_{\{n=q_k\}}. \quad (81)$$

This gives $t(n) = 0$ for $2 \leq n \leq 7$ and $t(8) = 1$. The selected earlier center other than the path predecessor is unique. Its distance is $\sqrt{2}$ at corners and 1 at side interiors.

Proof. At the corner $(0,0)$, the predecessor is $(-1,0)$ at distance 1. For $k \geq 3$, the unique next closest point in (80) is $(-1,1)$ at squared distance 2. Every remaining point has squared distance at least 4. Thus there is no corner tie to resolve.

At an interior point $(0,j)$ with $1 \leq j < \ell_k$, the predecessor is $(0,j-1)$. The unique rectangle point at distance 1 is $(-1,j)$. Every other previously placed point is farther after the predecessor is removed. These cases cover every index $n \geq 2$.

For $k \geq 6$, the corner table gives

$$C_{k-4} = C_k - z_k + v_k, \quad v_{k-4} = v_k, \quad \ell_{k-4} = \ell_k - 2. \quad (82)$$

Therefore the corner's selected center $(-1,1)$ has index q_{k-4} , and the interior selected center $(-1,j)$ has index $q_{k-4} + j - 1$. This remains valid at $j = \ell_k - 1$: it is then the next inner corner, since $j - 1 = \ell_{k-4}$. Finally,

$$q_k - q_{k-4} = 2k - 4, \quad (83)$$

so the two index formulas are precisely (81). Direct examination of $P_2, \dots, P_8$ verifies the remaining cases. $\square$

Corollary 13.3 (Delay properties). *The function t is nondecreasing, unbounded, and has increments 0 or 1. It takes every nonnegative integer value. For sufficiently large k ,*

$$t(q_k - 1) = t(q_k) = t(q_k + 1) = q_{k-4}. \quad (84)$$

Moreover $t(n) < n$ and

$$d(n) := n - t(n) = 4\sqrt{n} + \mathcal{O}(1). \quad (85)$$

The sequence a_n is positive for $n \geq 1$ and strictly increasing from a_7 onward.

Proof. The delay is $2k - 4$ at the initial corner and $2k - 3$ elsewhere on its side. The interior formula increases by 1 except for the first step after a corner; the side transition gives the other equality in (84). The initial plateau is explicit. Unboundedness and increments in $\{0, 1\}$ prove the range assertion. Equation (79) implies (85). Finally, $t(n) \geq 1$ for $n \geq 8$, so the recurrence adds a positive value at each such step. $\square$

The quarter-square plateau rule is already encoded by A265409 and its connection to A078510 [6, 7]. The closed expression above and the occupied-set proof make the geometric input to the asymptotics explicit; they do not introduce a new spiral model. The threefold corner use is also described in the existing implementation [10].

14 Constructing a useful normalizer

The following identities are exact, with derivatives taken in x :

$$w' = \frac{w}{2r^2(1+w)}, \quad (86)$$

$$p := \Phi' = \frac{w}{4r} = e^{-w}, \quad p' = -\frac{w^2}{8r^3(1+w)}, \quad (87)$$

$$s := \Psi' = \frac{w^2}{4r^2(1+w)} \left(\frac{1}{8} + \frac{1}{1+w} \right). \quad (88)$$

For example $dw/dr = w/[r(1+w)] \leq 1/r$. Direct differentiation therefore gives the uniform estimates

$$p = \mathcal{O}(w/r), \quad p' = \mathcal{O}(w/r^3), \quad p'' = \mathcal{O}(w/r^5), \quad s = \mathcal{O}(w/r^2), \quad s' = \mathcal{O}(w/r^4). \quad (89)$$

The formal leading balance for the recurrence is $p = e^{-dp}$ with $d \simeq 4r$. It gives $4rp = W_0(4r) = w$. Integrating the exact choice $p = w/(4r)$ yields Φ , up to an additive constant that can be absorbed in the final amplitude. This motivates the leading exponent, but it does not prove the theorem.

The order r^{-2} transport equation needs both the discrete backward-step term $-p^2/2$ and the side phase in the exact delay. We retain the former in the residual calculation below. It contributes to the $w^2/32$ part of Ψ ; replacing the backward difference formally by a derivative discards that transport contribution. No theorem about a continuous-delay surrogate is used here.

Let B_2 be the continuous periodic extension of

$$B_2(\theta) = \theta^2 - \theta + \frac{1}{6} \quad (0 \leq \theta \leq 1). \quad (90)$$

It is globally Lipschitz on the phase circle, although its derivative jumps at integers. Define

$$\theta(x) = \{2\sqrt{x}\}, \quad \Xi(x) = \frac{w^2}{16r} B_2(\theta(x)), \quad F(x) = H(x) + \Xi(x), \quad f_n = e^{F(n)}. \quad (91)$$

Here $\{\cdot\}$ denotes fractional part, and the values at wrap points are well defined by continuity. Uniformly,

$$\Xi(x) = \mathcal{O}(w^2/r) = o(1). \quad (92)$$

What the phase term means. The correction Ξ is a device for proving a summable residual. It is smaller than the final relative error $\mathcal{O}(w^3/r)$ in Theorem 12.1. Thus the theorem does not identify Ξ as a sharp first oscillatory term in $a_n/(Ce^{H(n)})$. Section 17 quantifies its genuine role at the level of the normalizer's residual.

15 Uniform residual estimates including corners

Choose a sufficiently large fixed starting index and let E be the set of indices

$$q_k \quad \text{and} \quad q_k + 1 \quad (93)$$

thereafter. Enlarging E by finitely many points has no effect. The real phase boundaries are $k^2/4$. If k is even the boundary equals q_k , whereas if k is odd it lies $1/4$ after q_k . Consequently E covers both every unit interval that crosses a phase boundary and every index where the corner delay formula needs separate treatment. There are $\mathcal{O}(1)$ elements of E in every radius interval $j \leq \sqrt{n} < j+1$.

Define the normalized defect

$$\mathcal{D}_n = \frac{f_n - f_{n-1} - f_{t(n)}}{f_n}. \quad (94)$$

Only large n with $t(n) \geq 1$ are used in this definition; no value of $F(0)$ is required.

Lemma 15.1 (Local cancellation). *Uniformly as $n \rightarrow \infty$,*

$$\mathcal{D}_n = \begin{cases} \mathcal{O}(w^3/r^3), & n \notin E, \\ \mathcal{O}(w^2/r^2), & n \in E. \end{cases} \quad (95)$$

Moreover $1 - \mathcal{D}_n > 0$ exactly.

Proof. Outside E we have $k(n) = \lfloor 2r \rfloor$ and, with $\theta = \{2r\}$,

$$d = n - t(n) = 4r + \delta, \quad \delta = -3 - 2\theta. \quad (96)$$

There is no phase wrap on $[n-1, n]$. Differentiating the phase on that interval gives

$$\Xi(n) - \Xi(n-1) = \frac{w^2}{16r^2}(2\theta - 1) + \mathcal{O}(w^2/r^3). \quad (97)$$

Indeed the derivative of the amplitude is $\mathcal{O}(w^2/r^3)$ and the second derivative on a phase-smooth interval is $\mathcal{O}(w^2/r^3)$. At any index, including E , periodic Lipschitz continuity still gives the bound $\Xi(n) - \Xi(n-1) = \mathcal{O}(w^2/r^2)$.

The delay interval has length $\mathcal{O}(r)$ and radius comparable with r . Taylor's theorem and (89) yield

$$\Phi(n) - \Phi(n-d) = dp - \frac{d^2}{2}p' + \mathcal{O}(w/r^2), \quad (98)$$

$$\Psi(n) - \Psi(n-d) = ds + \mathcal{O}(w/r^2). \quad (99)$$

The phase correction must be handled by endpoints rather than a smooth Taylor expansion across its corners. Since

$$\sqrt{n-d} = r - 2 + \mathcal{O}(1/r),$$

the endpoint phases differ by $4 + \mathcal{O}(1/r)$, and the endpoint amplitudes in (91) differ by $\mathcal{O}(w^2/r^2)$. Periodicity and Lipschitz continuity imply

$$\Xi(n) - \Xi(n-d) = \mathcal{O}(w^2/r^2). \quad (100)$$

This estimate remains valid when the interval crosses several phase boundaries.

Put $U = F(n) - F(n-1)$ and $V = F(n) - F(t(n))$. Equations (97)–(100) and (87) give, off E ,

$$U = p + s + \frac{w^2}{16r^2}(2\theta - 1) + \mathcal{O}(w^2/r^3), \quad (101)$$

$$V = w + \delta p + \frac{w^2}{r(1+w)} + 4rs + \mathcal{O}(w^2/r^2). \quad (102)$$

In particular $U = p + \mathcal{O}(w^2/r^2)$ and $V - w = \mathcal{O}(w/r)$. Expanding both exponentials, and using $e^{-w} = p$, gives

$$1 - e^{-U} = p + s + \frac{w^2}{16r^2}(2\theta - 1) - \frac{p^2}{2} + \mathcal{O}(w^3/r^3), \quad (103)$$

$$e^{-V} = p - \delta p^2 - \frac{w^3}{4r^2(1+w)} - ws + \mathcal{O}(w^3/r^3). \quad (104)$$

For example the quadratic exponential remainder in $V - w$ is multiplied by p , and is therefore $p\mathcal{O}(w^2/r^2) = \mathcal{O}(w^3/r^3)$. The replacement of U^2 by p^2 has the same error order.

Subtracting (104) from (103) yields

$$\mathcal{D}_n = (1+w)s + \left(2\theta - 1 + \delta - \frac{1}{2}\right) \frac{w^2}{16r^2} + \frac{w^3}{4r^2(1+w)} + \mathcal{O}(w^3/r^3). \quad (105)$$

By (96), the parenthesis is $-9/2$. The remaining exact identity is

$$\begin{aligned} (1+w)s + \frac{w^3}{4r^2(1+w)} &= \frac{w^2}{32r^2} + \frac{w^2}{4r^2(1+w)} + \frac{w^3}{4r^2(1+w)} \\ &= \frac{9w^2}{32r^2}. \end{aligned} \quad (106)$$

All displayed leading terms cancel. At indices in E , $d - 4r$ and the phase difference coefficients are still bounded. The same estimates without the cancellation give $\mathcal{O}(w^2/r^2)$. Finally, $1 - \mathcal{D}_n = (f_{n-1} + f_{t(n)})/f_n > 0$ by positivity. $\square$

Lemma 15.2 (Absolute tails and blocks). *The defects are absolutely summable, with*

$$\sum_{l>n} |\mathcal{D}_l| = \mathcal{O}(w_n^3/r_n). \quad (107)$$

A block beginning near n and having index length $4r_n + \mathcal{O}(1)$ has

$$\sum_{l \text{ in block}} |\mathcal{D}_l| = \mathcal{O}(w_n^3/r_n^2). \quad (108)$$

Proof. The ordinary-index tail is bounded by a constant times

$$\int_r^\infty \frac{w(u)^3}{u^2} du = \mathcal{O}(w(r)^3/r), \quad w(u) = W_0(4u).$$

For completeness, integration by parts bounds the integral by its boundary term plus $(3/w(r))$ times itself, because $w'(u) \leq 1/u$ and w is increasing. For $w(r) \geq 6$ this is at most twice the boundary term. Exceptional indices have only $\mathcal{O}(1)$ representatives per unit radius and contribute the smaller $\mathcal{O}(w^2/r)$ tail. On a block of the stated length there are $\mathcal{O}(r)$ ordinary indices and $\mathcal{O}(1)$ exceptions, with comparable r, w throughout. Their respective bounds are $\mathcal{O}(w^3/r^2)$ and $\mathcal{O}(w^2/r^2)$. $\square$

16 Positive bounds and convergence to one amplitude

Set $b_n = a_n/f_n$ for $n \geq 1$. Define

$$\alpha_n = \frac{f_{n-1}}{f_{n-1} + f_{t(n)}}, \quad \beta_n = 1 - \alpha_n. \quad (109)$$

For large n these are strictly positive and sum to 1. The recurrence becomes exactly

$$b_n = (1 - \mathcal{D}_n)(\alpha_n b_{n-1} + \beta_n b_{t(n)}). \quad (110)$$

Lemma 16.1 (Positive bounded normalization). *There are constants $0 < m \leq M < \infty$ such that $m \leq b_n \leq M$ for all positive n .*

Proof. Take N large enough that $t(n) \geq 1$ and $|\mathcal{D}_n| \leq 1/2$ for $n \geq N$. Let m_0, M_0 be the minimum and maximum of the finite positive set $b_1, \dots, b_{N-1}$. Induction in (110) gives

$$m_0 \prod_{l=N}^n (1 - |\mathcal{D}_l|) \leq b_n \leq M_0 \prod_{l=N}^n (1 + |\mathcal{D}_l|).$$

At an earlier parent index the upper product is smaller, and the lower product larger, than the one at the current index, so the induction is valid for both parents. By absolute summability the upper product has a finite limit and the lower product has a strictly positive limit. Incorporating the initial finite segment proves the assertion. $\square$

We can now rewrite (110) in additive form,

$$b_n = \alpha_n b_{n-1} + \beta_n b_{t(n)} + e_n, \quad |e_n| \leq M |\mathcal{D}_n|. \quad (111)$$

In particular $\sum |e_n| < \infty$. We next prove convergence by an explicit mixing estimate instead of inferring it from this summability.

Choose a sufficiently large N_0 and recursively define

$$N_{j+1} = \min\{n > N_j : t(n) = N_j\}, \quad r_j = \sqrt{N_j}, \quad w_j = W_0(4r_j). \quad (112)$$

Corollary 13.3 ensures existence. Since $t(N_{j+1}) = N_j$,

$$N_{j+1} - N_j = 4r_j + \mathcal{O}(1), \quad r_{j+1} - r_j = 2 + \mathcal{O}(1/r_j). \quad (113)$$

Indeed the first delay estimate initially gives $N_{j+1} - N_j = 4r_{j+1} + \mathcal{O}(1)$; solving the corresponding quadratic difference gives both forms in (113). In particular r_j is comparable with j .

Define the memory interval and its diameter by

$$I_j = [t(N_j), N_j] \cap \mathbb{Z}, \quad D_j = \max_{i \in I_j} b_i - \min_{i \in I_j} b_i. \quad (114)$$

The common-anchor weight is

$$\gamma_j = \prod_{l=N_j+1}^{N_{j+1}} \alpha_l. \quad (115)$$

Lemma 16.2 (Common-anchor contraction). *The memory diameters satisfy*

$$D_{j+1} \leq (1 - \gamma_j) D_j + 2 \sum_{N_j < l \leq N_{j+1}} |e_l|. \quad (116)$$

Moreover,

$$\gamma_j = e^{-w_j} (1 + \mathcal{O}(w_j^2/r_j)) = \frac{w_j}{4r_j} + \mathcal{O}(w_j^3/r_j^2). \quad (117)$$

Proof. For $N_j < n \leq N_{j+1}$, substitute (111) repeatedly until all ancestors have indices at most N_j . Monotonicity of t implies that none lies below $t(N_j)$. Thus b_n is a convex combination of values in I_j , plus an error of absolute value at most $\sum_{N_j < l \leq n} |e_l|$. The error bound follows by induction with convex weights; it does not add the same error with a total coefficient greater than 1.

The path using only immediate predecessors reaches b_{N_j} with weight $\prod_{l=N_j+1}^n \alpha_l \geq \gamma_j$. The endpoint $n = N_j$ has weight 1 at this same anchor. As

$$I_{j+1} = [N_j, N_{j+1}] \cap \mathbb{Z},$$

all new memory rows share mass at least γ_j on the single value b_{N_j} . Remove that common mass from each row. Their remaining masses are $1 - \gamma_j$, so their range is bounded by $(1 - \gamma_j)D_j$. Adding the two possible endpoint errors proves (116).

From (94) and (109), the product telescopes exactly:

$$\gamma_j = \frac{f_{N_j}}{f_{N_{j+1}}} \prod_{l=N_j+1}^{N_{j+1}} (1 - \mathcal{D}_l)^{-1}. \quad (118)$$

Integration of $H' = p + s$ over the block, using (113), gives its leading increment w_j . The endpoint estimate (92) is sufficient for the correction:

$$F(N_{j+1}) - F(N_j) = w_j + \mathcal{O}(w_j^2/r_j).$$

By (108), the logarithm of the product remaining in (118) is $\mathcal{O}(w_j^3/r_j^2)$. Exponentiation and $e^{-w_j} = w_j/(4r_j)$ prove (117). $\square$

Lemma 16.3 (Diameter rate). *We have $D_j = \mathcal{O}(w_j^2/r_j)$.*

Proof. The block defect bound reduces (116) to

$$D_{j+1} \leq (1 - \gamma_j)D_j + K \frac{w_j^3}{r_j^2} \quad (119)$$

for some fixed K . Use the positive barrier $B(r) = w(r)^2/r$. Its logarithmic derivative is

$$\frac{B'(r)}{B(r)} = \frac{1 - w}{(1 + w)r}. \quad (120)$$

Consequently $B(r_j) - B(r_{j+1}) = \mathcal{O}(w_j^2/r_j^2)$, whereas $\gamma_j B(r_j) \sim w_j^3/(4r_j^2)$. For large enough j there is $c > 0$ such that

$$\gamma_j B(r_j) - (B(r_j) - B(r_{j+1})) \geq c \frac{w_j^3}{r_j^2}.$$

Choose $A \geq K/c$ and also large enough to dominate the finite starting diameter. Induction in (119) gives $D_j \leq AB(r_j)$. $\square$

Proof of Theorem 12.1. The convex maximum principle applied to (111) puts every future value, $n > N_j$, in

$$\left[\min_{i \in I_j} b_i - T_j, \max_{i \in I_j} b_i + T_j \right], \quad T_j = \sum_{l > N_j} |e_l|. \quad (121)$$

This follows by induction: each new convex average stays between the previous extremes, and the accumulated absolute errors can enlarge them by at most their sum. Equations (107) and Lemma 16.3 show that the diameter of (121) is

$$D_j + 2T_j = \mathcal{O}(w_j^3/r_j) \longrightarrow 0.$$

Thus the full sequence b_n is Cauchy and converges to some C . Lemma 16.1 gives $0 < m \leq C \leq M < \infty$.

If $N_j \leq n \leq N_{j+1}$, comparison with (121), including its limiting value, yields

$$b_n = C + \mathcal{O}(w_n^3/r_n).$$

Since $e^{\Xi(n)} = 1 + \mathcal{O}(w_n^2/r_n)$ and $C > 0$, multiplying by $f_n = e^{H(n) + \Xi(n)}$ proves (71). Finally $w \sim \log(4r)$, so $\Phi \sim r \log r/2$ and $\Psi = o(\Phi)$; this proves (72). $\square$

17 Two refinements of the normalization argument

These refinements quantify features of the proof. Neither supplies a sharper relative expansion of a_n .

17.1 Absolute defect of the smooth normalizer

Use $e^{H(n)}$ without Ξ and define

$$\mathcal{D}_n^H = \frac{e^{H(n)} - e^{H(n-1)} - e^{H(t(n))}}{e^{H(n)}}. \quad (122)$$

The same calculation as in Lemma 15.1, now omitting the phase derivative, gives

$$\mathcal{D}_n^H = \frac{w^2}{16r^2}(1 - 2\{2r\}) + \mathcal{O}(w^3/r^3) \quad (n \notin E), \quad (123)$$

and $\mathcal{D}_n^H = \mathcal{O}(w^2/r^2)$ on E .

Proposition 17.1 (Nonsummable smooth residual). *For any fixed sufficiently large lower limit n_* , a finite constant K_* exists such that*

$$\sum_{n=n_*}^N |\mathcal{D}_n^H| = \frac{w_N^3}{48} + \frac{w_N^2}{32} + K_* + \mathcal{O}(w_N^3/r_N). \quad (124)$$

In particular the smooth normalizer has non-absolutely-summable defects.

Proof. Let $g(\theta) = |1 - 2\theta|$ on $[0, 1]$, continuously and periodically extended, and set

$$h(x) = \frac{w(x)^2}{16x} g(\{2\sqrt{x}\}).$$

The inequality $||u| - |v|| \leq |u - v|$, (123), and the exceptional-index estimates show that $\sum(|\mathcal{D}_n^H| - h(n))$ converges, with tail $\mathcal{O}(w_N^3/r_N)$. The function h is continuous and piecewise differentiable. Its derivative is absolutely integrable on a tail, and

$$\int_N^\infty |h'(x)| dx = \mathcal{O}(w_N^2/r_N).$$

For example differentiating the phase contributes $\mathcal{O}(w^2/r^3)$ to h' ; integrating with $dx = 2r dr$ gives the stated order, and differentiating its amplitude gives a smaller term. Comparing the integral on each unit interval with its endpoint value now proves that $\sum h(n) - \int h(x) dx$ has a finite limit with the same tail order.

With $x = r^2$, the integral becomes

$$\int \frac{w(r)^2}{8r} g(\{2r\}) dr.$$

The mean of g is $1/2$. Its zero-mean part has a bounded periodic primitive, so integration by parts against $w^2/(8r)$ gives a convergent contribution with tail $\mathcal{O}(w^2/r)$. The mean part integrates exactly, because $dr/r = (1 + w) dw/w$:

$$\frac{1}{16} \int \frac{w^2}{r} dr = \frac{1}{16} \int w(1 + w) dw = \frac{w^3}{48} + \frac{w^2}{32}.$$

Combining the convergent differences proves (124). $\square$

Thus $\Xi = o(1)$ can be negligible at the scale of the final equivalent while remaining indispensable to this particular absolute-defect proof. The constant K_* in (124) depends on the lower limit and is unrelated to the amplitude C .

17.2 The homogeneous contraction product

For fixed sufficiently large J , define

$$P_{J,j} = \prod_{h=J}^{j-1} (1 - \gamma_h). \quad (125)$$

This is the product controlling the unforced diameter inequality; it is not a representation of C .

Proposition 17.2 (Quantitative contraction product). *There is a constant $K_J > 0$ such that*

$$P_{J,j} = K_J \exp\left(-\frac{w_j^2}{16} - \frac{w_j}{8}\right) \left(1 + \mathcal{O}(w_j^3/r_j)\right). \quad (126)$$

In particular this product is smaller than every fixed inverse power of r_j eventually, and $\sum_j \gamma_j = \infty$.

Proof. Set $G(r) = w(r)^2/16 + w(r)/8$. Its derivative is exactly $G'(r) = w(r)/(8r)$. Equation (113) gives

$$G(r_{j+1}) - G(r_j) = \frac{w_j}{4r_j} + \mathcal{O}(w_j/r_j^2).$$

Using (117) and the quadratic error in $\log(1 - \gamma_j)$, we obtain

$$\log(1 - \gamma_j) + G(r_{j+1}) - G(r_j) = \mathcal{O}(w_j^3/r_j^2).$$

Because r_j is comparable with j , this error is summable with tail $\mathcal{O}(w_j^3/r_j)$. Summing and exponentiating proves (126), with a positive finite K_J . The leading estimate $\gamma_j \sim w_j/(4r_j)$ also shows divergence of its sum. Finally $w_j \sim \log r_j$, so the quadratic exponential in (126) beats every fixed power. $\square$

The forced error in (119) explains why this very rapid homogeneous contraction does not by itself yield a comparably small remainder for $b_n - C$.

18 Threshold inversion with an absolute error

By Corollary 13.3, linear interpolation of $\log a_n$ from $n = 7$ onward gives a continuous strictly increasing function $\mathcal{A}(x)$. Denote its inverse by $X(L)$ for $L \geq 0$. Then, for $Y > 1$,

$$N(Y) = \lceil X(\log Y) \rceil. \quad (127)$$

This identity follows from the definition of interpolation. It does not justify taking the ceiling of an approximation to X .

[write] **7 October 2026.** This identity is item (1) of the transseries volume's `p0:thm:staircase`, with $A_n = \log a_n$ for $n \geq 7$ and the admissible interpolation $\mathcal{A}$; the estimate of Theorem 12.2 is this Part's own (front matter, "The inverses and the transseries volume").

The derivative of $e^w(w^2 - w + 1)$ is $e^w w(w + 1)$, strictly positive for $w > 0$. Thus (74) has exactly one positive solution for $L > 1/8$. Substituting $r_0 = w_0 e^{w_0}/4$ in (68) shows that

$$\Phi(r_0^2) = \frac{e^{w_0}}{8}(w_0^2 - w_0 + 1) = L. \quad (128)$$

Proof of Theorem 12.2. Let

$$Q_0 = \Psi(r_0^2) + \log C, \quad x_0 = r_0^2 - \frac{4r_0}{w_0}Q_0. \quad (129)$$

Since $Q_0 = \mathcal{O}(w_0^2)$, the shift is $x_0 - r_0^2 = \mathcal{O}(r_0 w_0) = o(r_0^2)$. Its linear contribution to Φ is exactly $-Q_0$. The quadratic remainder is

$$\mathcal{O}\left((r_0 w_0)^2 \frac{w_0}{r_0^3}\right) = \mathcal{O}(w_0^3/r_0),$$

and the change in Ψ is $\mathcal{O}(w_0^2/r_0)$ by (88). Therefore

$$H(x_0) + \log C = L + \mathcal{O}(w_0^3/r_0). \quad (130)$$

Theorem 12.1 gives $\log a_n = H(n) + \log C + \mathcal{O}(w_n^3/r_n)$. Convex interpolation preserves this endpoint error. Moreover $H'' = \mathcal{O}(w/r^3)$, so its own interpolation error on a unit interval is $\mathcal{O}(w/r^3)$. Hence, uniformly for real $x \rightarrow \infty$,

$$\mathcal{A}(x) = H(x) + \log C + \mathcal{O}(w^3/r). \quad (131)$$

Near x_0 , H' is comparable with w_0/r_0 . Evaluating (131) at $x_0 \pm Aw_0^2$, with sufficiently large fixed A , makes the resulting change of H dominate both errors in (130) and (131). Monotonicity brackets $X(L)$ between these points. Thus $X(L) = x_0 + \mathcal{O}(w_0^2)$, and the ceiling in (127) changes this by at most 1. This proves (75).

Finally $\Phi(r^2) \sim r \log r/2$ implies $r_0 \sim 2L/\log L$. The correction in (75) is $o(r_0^2)$, proving (76) with coefficient 4. $\square$

Equivalently, solving $H(x) + \log C = L$ implicitly has absolute horizontal error $\mathcal{O}(w^2)$. This is the familiar conversion of a vertical residual into a root error by division by a positive local slope. The contribution specific to the present sequence is the justified residual size and the exact form of H .

19 Every fixed inverse logarithmic order

A finite inverse-logarithmic truncation is less precise in absolute terms than (75), but it exposes the successive logarithmic scales without the unknown amplitude. Define

$$V = 2W_0(\sqrt{2L}), \quad u = V^{-1}. \quad (132)$$

Then $V^2 e^V = 8L$. The exact inverse of Φ can be expressed through one analytic germ.

Lemma 19.1 (Analytic inverse germ). *There is a unique analytic function $\delta(u)$ near $u = 0$, with $\delta(0) = 0$, satisfying*

$$\delta + \log Q(u, \delta) = 0, \quad (133)$$

$$Q(u, \delta) = (1 + u\delta)^2 - u(1 + u\delta) + u^2. \quad (134)$$

The logarithm is the analytic branch taking value 0 at $Q = 1$. For all sufficiently large L ,

$$w_0 = V + \delta(1/V). \quad (135)$$

Proof. The left side of (133) is analytic near $(u, \delta) = (0, 0)$, vanishes there, and has derivative 1 with respect to δ there. The analytic implicit-function theorem supplies a unique convergent germ. Exponentiating (133) gives

$$e^{\delta}Q(u, \delta) = 1. \quad (136)$$

Writing $w_0 = V + \delta$ and dividing (74) by $V^2e^V = 8L$ yields exactly (136). For small positive u , $V + \delta(u) > 0$; the strict monotonicity established above identifies it with the positive solution. $\square$

The first six coefficients are

$$\delta(u) = u - \frac{5}{2}u^2 + \frac{10}{3}u^3 - \frac{29}{12}u^4 - \frac{14}{5}u^5 + \frac{403}{30}u^6 + \mathcal{O}(u^7). \quad (137)$$

Define the analytic germ

[write] **7 October 2026.** As a formal series, δ is the solution U of the transseries volume's `p0:thm:lambert-centered` with $a = 1$, $b = 2$, $Q(s) = \log(1 - s + s^2)$ and $t = u$, and $V = 2W_0(\sqrt{2L})$ is the root given by its `p0:thm:lambert-core` for $V + 2\log V = \log(8L)$; the convergence of the germ is this Part's own (front matter, "The inverses and the transseries volume").

$$\mathcal{R}(u) = \frac{(1 + u\delta(u))^2}{Q(u, \delta(u))^2} = (1 + u\delta(u))^2 e^{2\delta(u)}. \quad (138)$$

Since $r_0 = 2Lw_0/(w_0^2 - w_0 + 1)$, we have the exact identity

$$r_0^2 = \frac{4L^2}{V^2} \mathcal{R}(1/V). \quad (139)$$

Its first six nonconstant coefficients are

$$\mathcal{R}(u) = 1 + 2u - u^2 - 3u^3 + \frac{10}{3}u^4 - \frac{1}{6}u^5 - \frac{389}{60}u^6 + \mathcal{O}(u^7). \quad (140)$$

Theorem 19.2 (Fixed inverse-logarithmic hierarchy). *Write $\mathcal{R}(u) = \sum_{j \geq 0} R_j u^j$ near 0. For every fixed integer $J \geq 0$,*

$$N(Y) = \frac{4L^2}{V^2} \left(\sum_{j=0}^J R_j V^{-j} + \mathcal{O}_J(V^{-J-1}) \right), \quad L = \log Y, \quad V = 2W_0(\sqrt{2L}). \quad (141)$$

The coefficients are exactly computable rational numbers. This statement concerns every fixed truncation order as $Y \rightarrow \infty$.

Proof. Analyticity gives the Taylor remainder $\mathcal{O}_J(u^{J+1})$ for each fixed J . By (75), $N(Y) - r_0^2 = \mathcal{O}(r_0 w_0) = \mathcal{O}(L)$, because $r_0 w_0 = 2Lw_0^2/(w_0^2 - w_0 + 1)$. Relative to $4L^2/V^2$, this is $\mathcal{O}(V^2/L)$, smaller than every fixed inverse power of V since $L = V^2 e^V/8$. Combining this with (139) proves the formula. $\square$

19.1 An exact coefficient generator

No symbolic algebra package is needed. Let $[u^m]$ denote coefficient extraction. Suppose $d_1, \dots, d_{m-1}$ are known and set $D_{m-1}(u) = \sum_{i=1}^{m-1} d_i u^i$. Then

$$d_m = -[u^m] \left(e^{D_{m-1}(u)} Q(u, D_{m-1}(u)) - 1 \right). \quad (142)$$

Indeed the as-yet unknown coefficient d_m enters the coefficient of degree m in $e^\delta Q - 1$ with coefficient 1: its occurrence through $u\delta$ is delayed to higher degree. This proves the triangular recursion and rationality. Having computed δ to the desired degree, (138) supplies R_j by finite truncated multiplication.

For an explicit rational implementation of an exponential, if $D(u) = \sum_{i \geq 1} d_i u^i$ and $E(u) = e^{D(u)} = \sum_{m \geq 0} e_m u^m$, then

$$e_0 = 1, \quad m e_m = \sum_{i=1}^m i d_i e_{m-i}. \quad (143)$$

Thus every operation in (142) is on rational numbers. One can separately certify a computed list by checking $e^\delta Q - 1 = 0$ to the desired degree and then checking

$$\mathcal{R}Q^2 - (1 + u\delta)^2 = 0 \quad (144)$$

with independent polynomial arithmetic. Such finite checks certify coefficients, not the analytic convergence of the germ; the latter was proved in Lemma 19.1.

Separation of precision claims. The germ itself has a positive convergence radius near $u = 0$. Formula (141) is nonetheless stated as a fixed-order asymptotic expansion for the actual integer threshold, because it omits the much smaller Newton/amplitude correction. No expansion of all relative orders for a_n follows. Conversely, retaining Ψ and $\log C$ in (75) gives much finer absolute precision than any fixed truncation of (141).

20 Reproducible finite checks

The companion archive contains the authored manuscript, bounded attributed sequence fixtures, exact verification code, optional numerical diagnostics, and offline build tools. The mandatory verification path uses Python's standard library only. It runs in ordinary and optimized modes, using explicit exceptions rather than removable assertions, and requires byte-identical canonical receipts.

The exact checks cover the following distinct questions:

1. An independent unit-step coordinate construction checks the closed delayed index through $n = 250000$, including uniqueness and squared-distance class of the selected earlier center. Its local lattice search is exact because the successful candidate lies at squared distance 1 or 2, and no nonzero integer squared distance lies between these values.
2. A separate global search compares every earlier point at all indices $2 \leq n \leq 4096$. Complete early pre-corner occupied rectangles are compared as sets. These tests supplement the all-index rectangle proof.
3. Exact integer recurrence evaluation reproduces all 64 supplied A078510 values at indices 0 through 63. Only this small attributed fixture is redistributed; no full OEIS export is included.
4. Exact rational generation and independent coefficient identities verify the inverse germs, including the six displayed coefficients. The analytic implicit-function proof supplies the all-fixed-order assertion.
5. Type and domain guards, fixture corruption checks, closed archive inventories, manifest hashes, and output no-clobber tests exercise the verification and release machinery itself.

The documented full replay and build commands, precise check counts, and output safeguards are in the archive’s reproducibility guide. A clean build produces the PDF, its matching TeX source, the code archive, and a release manifest. The PDF is compiled with fixed metadata and strict rejection of TeX warnings. The same four outputs must be byte-identical in two clean extracted builds under the same toolchain.

Floating-point exploration is separate from these mandatory exact checks. Optional high-precision residual samples can illuminate the cancellation scales, and long logarithmic recurrence runs can suggest a numerical amplitude. Neither supplies a uniform error proof, a certified value of C , an effective onset, or a proof by sampling. No decimal digits of C are certified in this report.

[write] 7 October 2026. In this report the package’s programs are `code/191-square-*.py`, its fixtures and recorded outputs `data/191-square-*`, and its `README_REPRODUCIBILITY.md`, `DATA_SOURCES.md`, `code/README.md` and `data/README.md` are `191-square-*.md`. `Report191.tex` (its text is this Part), the delivered PDF, the package `README.md` and `SHA256SUMS.json` are not shipped; the replay and build scripts need the delivered inventory. The report’s `README` gives a route that runs the mandatory checker on a copy.

21 Source attribution and the bounds of the comparison

The sources distinguish the established model and general techniques from the recurrence-specific theorem proved here.

21.1 The prior sequence and geometric rule

Fernandez’s account of Spiro-Fibonacci sequences [5] defines the closest-two-earlier-values square construction, gives sample terms and difference patterns, and discusses alternative seeds. Its inspected text supplies no growth theorem. The OEIS record A078510 [6] gives the delayed recurrence through A265409; it credits that recurrence connection to Antti Karttunen in December 2015. The quarter-square predecessor encoding is therefore explicitly prior work [7]. The inspected related records A265370 and A265404 concern binary length and summand counts and do not state the full equivalent proved above [8, 9].

Ryde’s version-75 `Math::NumSeq::SpiroFibonacci` implementation and embedded documentation [10] were read as source text. They implement side counters and a queue, explain the repeated inward value at corners, and discuss seeds. They do not give a growth formula. That program was not executed, and its full source is not redistributed in the companion archive.

The metric matters. Our exact theorem uses Euclidean distances between cell centers, which gives the unique selection proved in Proposition 13.2. A different metric or literal distance between closed squares can create ties and does not alone define this same sequence.

21.2 Related delay theory and its hypotheses

Kundrát’s 2005 paper [11], whose full text was inspected, studies a discretized pantograph equation of the form

$$\Delta x_n = -ahx_n + bhx_{\tau_n}.$$

Its discrete comparison theorem assumes $0 < ah < 1$. The present recurrence has $a = 0$ in that notation, so that result cannot be applied directly. The paper’s continuous comparison theorem also has a decreasing-derivative condition on the delayed argument, whereas the derivative of $t - 4\sqrt{t}$ increases.

The 2006 proportional-delay paper [12], also inspected in full, assumes a fixed ratio $0 < \lambda < 1$ and a strict contraction condition $|1 + a_{ii}| < 1$ on its nondelayed matrix coefficient. Here $t(n)/n \rightarrow 1$ and the nondelayed coefficient in the difference equation is zero, violating those hypotheses. These papers are relevant comparison methodology, rather than ready-made proofs of Theorem 12.1.

Appleby and Buckwar [13] develop constructive comparisons for functional differential equations with unbounded delay. The publisher’s full text was unavailable, but selected theorem statements were subsequently inspected in a publicly readable OCR reproduction, with the usual transcription limitations. Their displayed model is the continuous equation $x'(t) = -ax(t) + bx(t - \tau(t))$. Theorem 4 assumes $a > b > 0$, while Theorem 8 and the growth case of Theorem 9 assume $b > a > 0$. These positive-damping continuous statements do not directly establish the present zero-damping discrete equivalent. Theorem 8 gives a logarithmic-rate limit normalized by an integral of $1/\sigma$, while Theorem 9 uses an integral of $1/\tau$; neither of these conclusions is the amplitude equivalent proved here. This selected-statement comparison does not exclude every result in that article or replace a complete literature review.

21.3 Existing Lambert and inverse methods

Lambert characteristic balance, WKB-style logarithmic normalization, variation of constants, positive comparison, and analytic implicit inversion are established tools. The relevant existing Lambert guide [14] treats exponential modes of constant-delay equations through W ; this motivates a local balance but does not by itself handle the present variable delay, corner phases, or normalized convergence. Existing transseries material [15, 16] also treats Lambert logarithmic expansions, formal versus analytic realization, and residual-to-root transport. The use of these mechanisms in Sections 14 and 18 is not a claim to a new general method.

The supplied repository comparison was pinned to revision

9ba11eaf7c1b6a27ccfc0bab63e22f97768cb704.

The inspected constant-delay and inversion statements, rather than absence of a sequence-name search hit, support this distinction. Inspection was bounded to relevant sections; it does not certify what might occur in uninspected files or later revisions.

[write] 7 October 2026. The pinned commit exists (3 October 2026) and contains both cited files. The “supplied manuscript” *Transseries for Mere Mortals* is one of the four tutorials merged into the transseries volume’s Part Q0 (label `q0:sec:why`).

The earlier hexagonal report [17] concerns a different recurrence containing the additional path term a_{n-2} and further neighbors. Its scale is $C\phi^n$, where $\phi = (1 + \sqrt{5})/2$. The present closest-two square rule lacks that term and has the subexponential scale in (71). Its proof cannot be obtained by changing a geometric constant in the hexagonal theorem.

[write] 7 October 2026. Report190 is Part I of this report. The two Parts share the construction family and the method, as this paragraph says, and no theorem; the Guide’s table sets the two side by side.

Claim boundary. No inspected passage states the particular full equivalent proved here. This is a bounded source finding, not an exhaustive novelty clearance. The contribution presented is the explicit square-rule analysis: transport including the discrete term, phase-corrected summable defects, genuine convergence to a positive amplitude, uniform quantitative error, and the ensuing sequence-specific inverse formulas. The methods and prior construction retain the attributions above.

22 Limitations and further questions

The following problems remain open within this report’s scope.

1. *Certified amplitude.* Construct computable two-sided bounds for C with explicit finite cutoffs and constants. The present positive-product argument proves $C > 0$ but does not certify decimal digits or a practical convergence threshold.
2. *Sharper relative error.* The rate $\mathcal{O}(w^3/r)$ comes from an absolute tail of residuals. Additional signed cancellation or a higher transport normalization may improve it. Such an improvement is needed before treating $\Xi = \mathcal{O}(w^2/r)$ as a resolved sequence-level phase correction.
3. *Further sequence corrections.* Determine whether there is a rigorous relative hierarchy with all side and corner layers, and what scale and phase structure it uses. The inverse-logarithmic germ proved here does not answer this question.
4. *Effective threshold recovery.* Combine certified amplitude bounds, explicit remainder constants, and finite exact recurrence checks to obtain practical integer-threshold certificates. Equation (127) alone supplies no rounding guarantee for an approximate inverse.
5. *General delayed models and source closure.* Isolate assumptions on near-identity delays and phase geometry that reproduce the common-anchor argument, and compare with the unbounded-delay literature beyond the selected statements inspected here. Such a generalization would require its own proof and source review.

These limitations do not weaken the full equivalent or the error estimates proved here; they separate those established statements from stronger numerical, asymptotic, or priority claims.

[write] **7 October 2026.** These five questions are items 5–9 of Section 23; Part I answers none of them.

23 Further questions and research

[Added 7 October 2026, batch 110.] This section gathers, under Vladimir’s standing rule of 4 October 2026, every question and every claim left unproved by the two manuscripts. Report 190 has no questions section; its limits are stated in its Scope paragraph, in Section 6.1, after Theorems 7.1 and 9.2, and at the end of Section 11; the write states them as questions (items 1–4). Report 191’s Section 22 lists five (items 5–9). The delivered texts stay where they are. The two lists are kept apart because the Parts share no theorem; no question of one Part is answered by the other. No claim of either manuscript was found to be false.

Part I (hexagonal spiral).

1. *Growing order.* Theorem 1.1 holds for every fixed J ; it is “not a claim of convergence as $J \rightarrow \infty$ ”, and the factors $Q^{-k(k-1)/2}$ in the profile constants γ_k (Theorem 7.1) show that the proof gives no control uniform in k . Does $\sum_k F_k$ converge, or is it only asymptotic, and what is the optimal truncation? Open.
2. *The amplitudes.* No closed elementary expression for C_0 or C_1 is claimed (Section 11). Is either expressible in known constants? The certificates of Theorem 4.1 give 120 digits for any such test. Open.
3. *Effective onsets and certified inverse constants.* The constants K_J and onsets of Theorem 9.2 and the error constants of Theorem 1.1 are existence statements. Section 6.1 describes an

interval algorithm for each fixed $F_k(n)$ with explicit norm bounds, but the package does not implement it, and no effective inverse onset is given. Open.

4. *Other seeds.* Part I notes that the amplitude argument “works for nonnegative a_0 and positive a_1 ” and proves seed independence beyond every fixed order for the two OEIS seeds (Corollary 6.2); a theorem uniform in a seed parameter is not asserted (Section 11). Open.

Part II (square spiral).

5. *Certified amplitude* (Report 191, Section 22, item 1). Two-sided computable bounds for C with explicit cutoffs; the positive-product argument proves $C > 0$ and no digits. Open.
6. *Sharper relative error* (item 2). The rate $O(w^3/r)$ comes from an absolute tail of residuals; signed cancellation or a higher transport normalization might improve it, which is needed before $\Xi = O(w^2/r)$ can be read as a phase correction of the sequence (Section 14, “What the phase term means”). Open.
7. *Further sequence corrections* (item 3). A rigorous relative hierarchy with all side and corner layers, its scales and phases. The inverse-logarithmic hierarchy of Theorem 19.2 does not answer it. Open.
8. *Effective threshold recovery* (item 4). Certified amplitude bounds, explicit remainder constants and finite exact checks combined into integer-threshold certificates; (127) alone gives no rounding guarantee for an approximate inverse. Open.
9. *General delayed models and source closure* (item 5). Assumptions on near-identity delays and phase geometry that reproduce the common-anchor argument, and a comparison with the unbounded-delay literature beyond the statements Report 191 inspected (Section 21). Open.

A question common to both Parts, which neither manuscript states, is whether one framework covers spiral-Fibonacci rules on other lattices and with other neighbour sets; the two proofs share an outlook (exact delay reduction, normalization, inverse), not a theorem, and nothing is proved about such a framework.

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- [15] *Transseries for Mere Mortals*, supplied manuscript, relevant Lambert and WKB sections. Source comparison limited to the inspected sections.

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